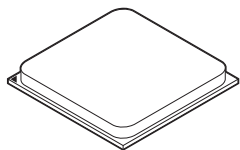


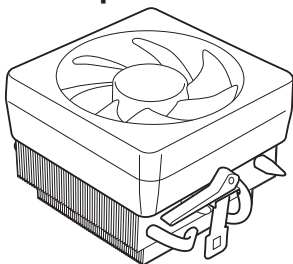
Quick Start

Thank you for purchasing the MSI® **B550M PRO-VDH WIFI/ B550M PRO-VDH WIFI6** motherboard. This Quick Start section provides demonstration diagrams about how to install your computer. Some of the installations also provide video demonstrations. Please link to the URL to watch it with the web browser on your phone or tablet. You may have even link to the URL by scanning the QR code.

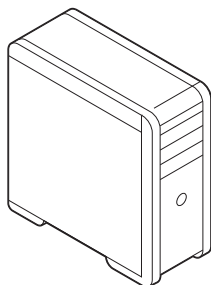
Preparing Tools and Components



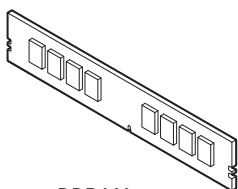
AMD® AM4 CPU



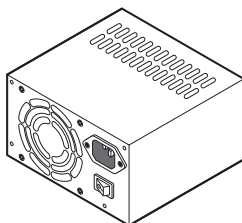
CPU Fan



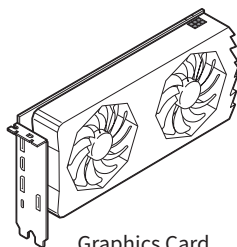
Chassis



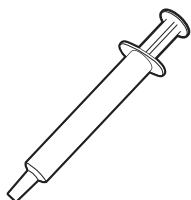
DDR4 Memory



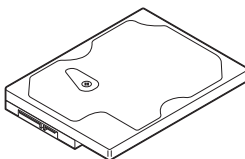
Power Supply Unit



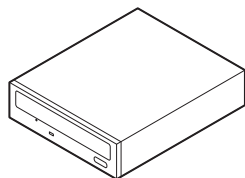
Graphics Card



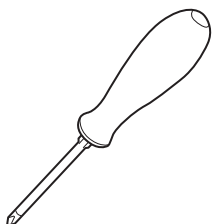
Thermal Paste



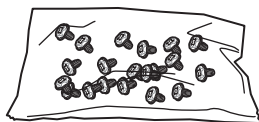
SATA Hard Disk Drive



SATA DVD Drive



Phillips Screwdriver



A Package of Screws

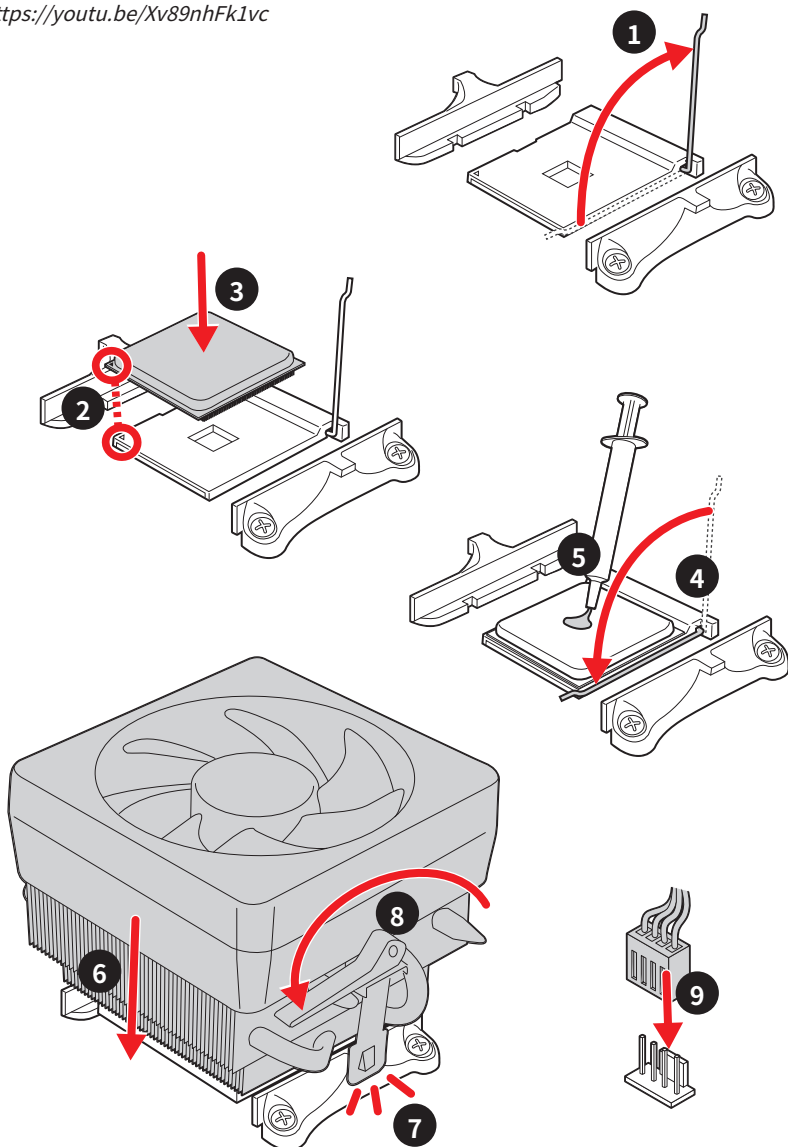
Safety Information

- The components included in this package are prone to damage from electrostatic discharge (ESD). Please adhere to the following instructions to ensure successful computer assembly.
- Ensure that all components are securely connected. Loose connections may cause the computer to not recognize a component or fail to start.
- Hold the motherboard by the edges to avoid touching sensitive components.
- It is recommended to wear an electrostatic discharge (ESD) wrist strap when handling the motherboard to prevent electrostatic damage. If an ESD wrist strap is not available, discharge yourself of static electricity by touching another metal object before handling the motherboard.
- Store the motherboard in an electrostatic shielding container or on an anti-static pad whenever the motherboard is not installed.
- Before turning on the computer, ensure that there are no loose screws or metal components on the motherboard or anywhere within the computer case.
- Do not boot the computer before installation is completed. This could cause permanent damage to the components as well as injury to the user.
- If you need help during any installation step, please consult a certified computer technician.
- Always turn off the power supply and unplug the power cord from the power outlet before installing or removing any computer component.
- Keep this user guide for future reference.
- Keep this motherboard away from humidity.
- Make sure that your electrical outlet provides the same voltage as is indicated on the PSU, before connecting the PSU to the electrical outlet.
- Place the power cord such a way that people can not step on it. Do not place anything over the power cord.
- All cautions and warnings on the motherboard should be noted.
- If any of the following situations arises, get the motherboard checked by service personnel:
 - Liquid has penetrated into the computer.
 - The motherboard has been exposed to moisture.
 - The motherboard does not work well or you can not get it work according to user guide.
 - The motherboard has been dropped and damaged.
 - The motherboard has obvious sign of breakage.
- Do not leave this motherboard in an environment above 60°C (140°F), it may damage the motherboard.

Installing a Processor

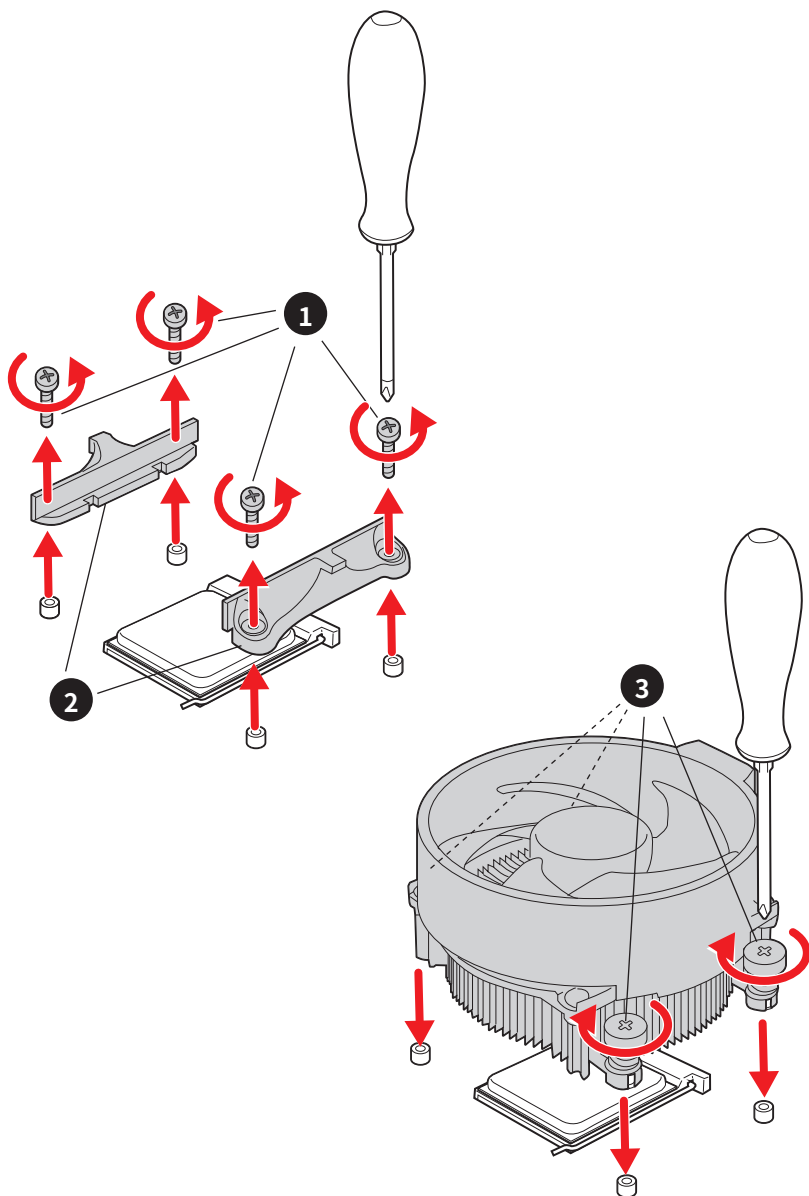


<https://youtu.be/Xv89nhFk1vc>



 Important

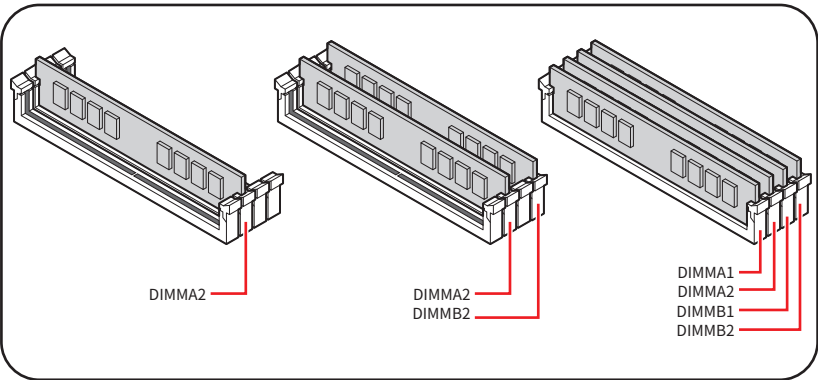
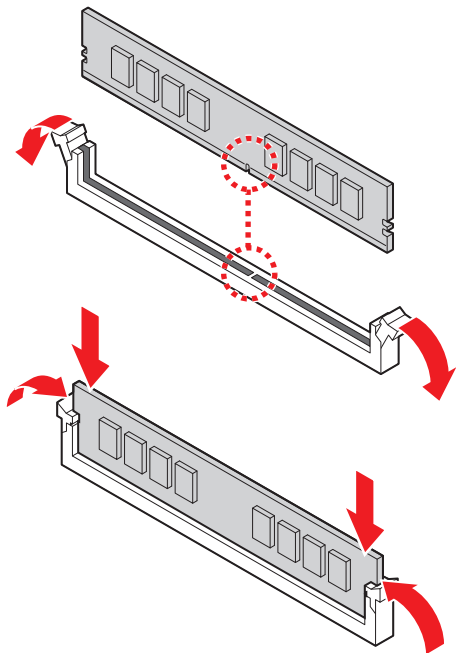
If you are installing the screw-type CPU heatsink, please follow the figure below to remove the retention module first and then install the heatsink.



Installing DDR4 memory



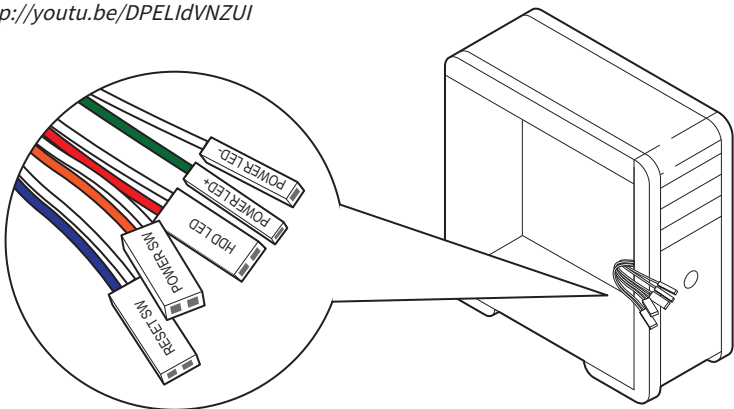
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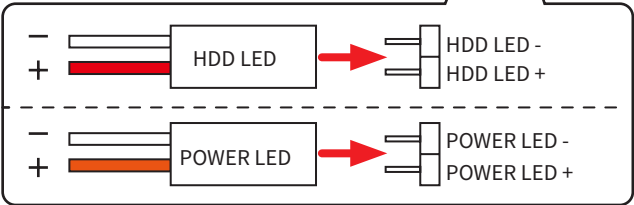
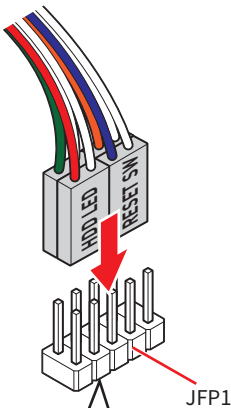
Connecting the Front Panel Header



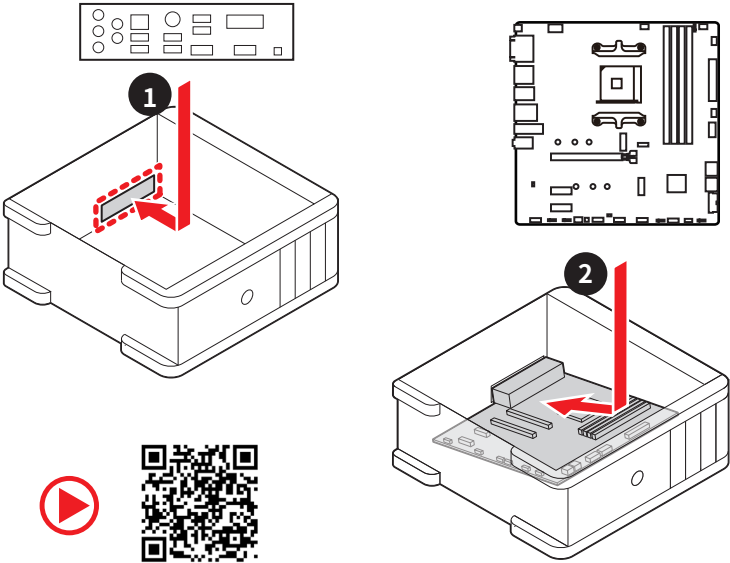
<http://youtu.be/DPELIdVNZUI>



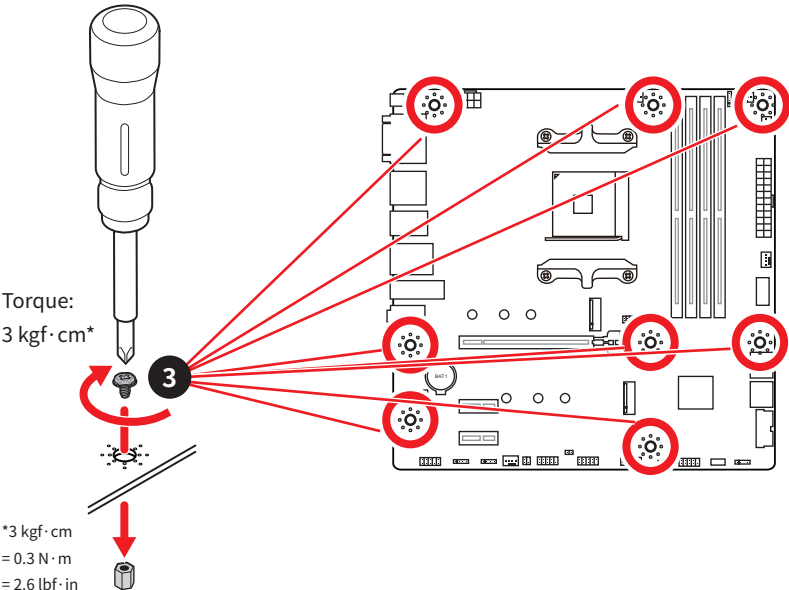
Power LED		Power Switch	
JFP1		Reserved	
HDD LED		Reset Switch	
1	HDD LED +	2	Power LED +
3	HDD LED -	4	Power LED -
5	Reset Switch	6	Power Switch
7	Reset Switch	8	Power Switch
9	Reserved	10	No Pin



Installing the Motherboard



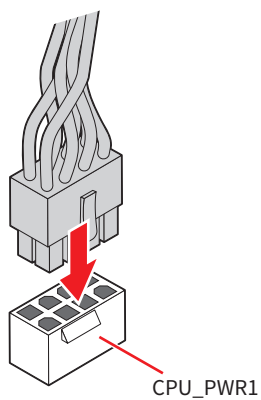
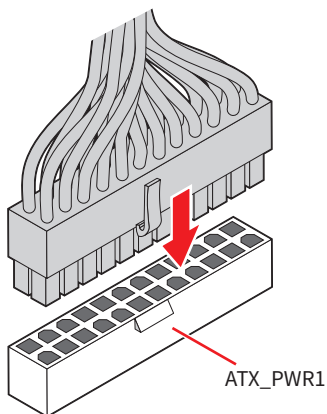
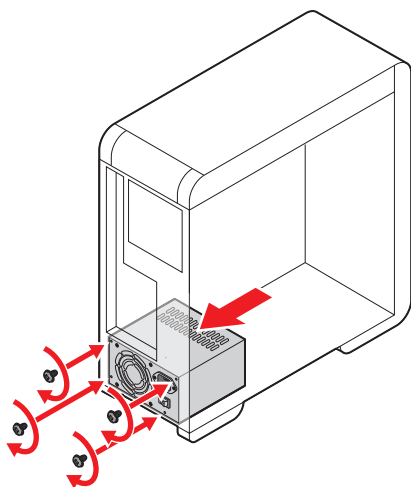
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Connecting the Power Connectors



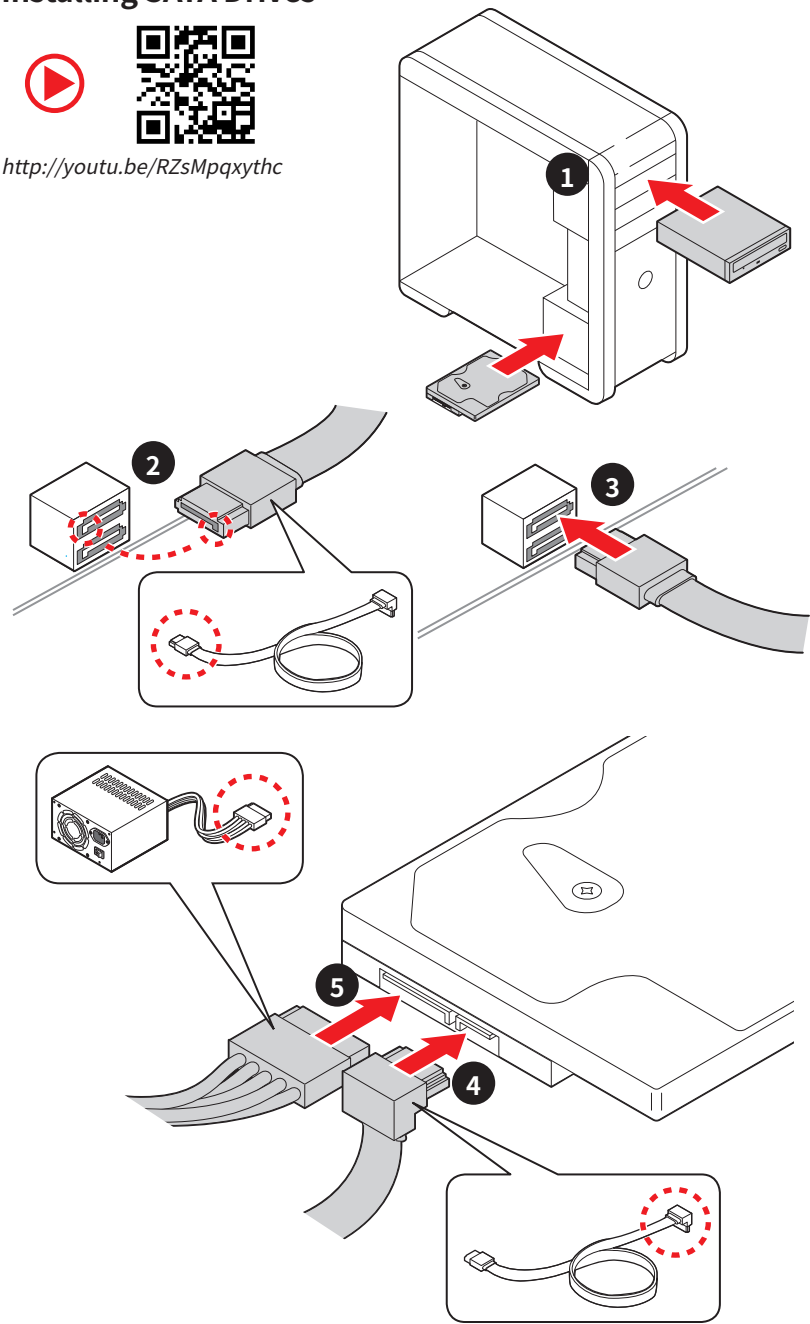
http://youtu.be/gkDYyR_83I4



Installing SATA Drives



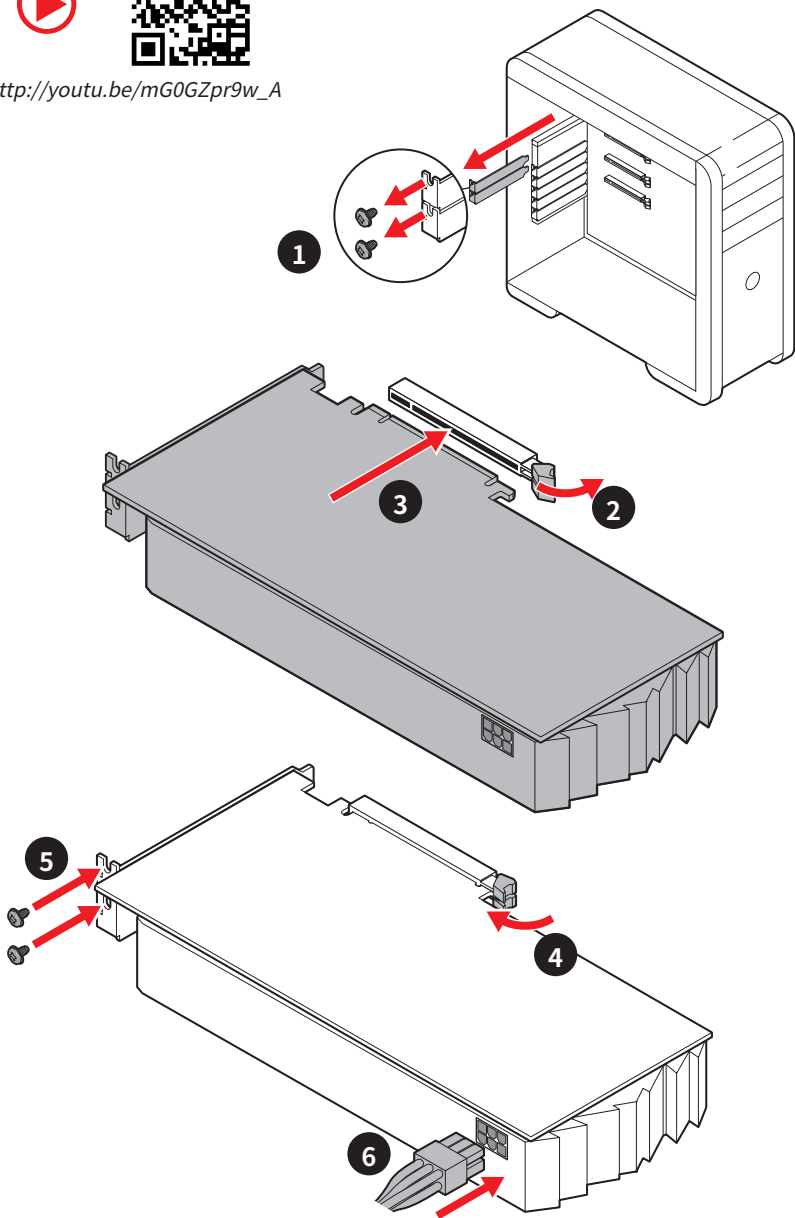
<http://youtu.be/RZsMpqxythc>



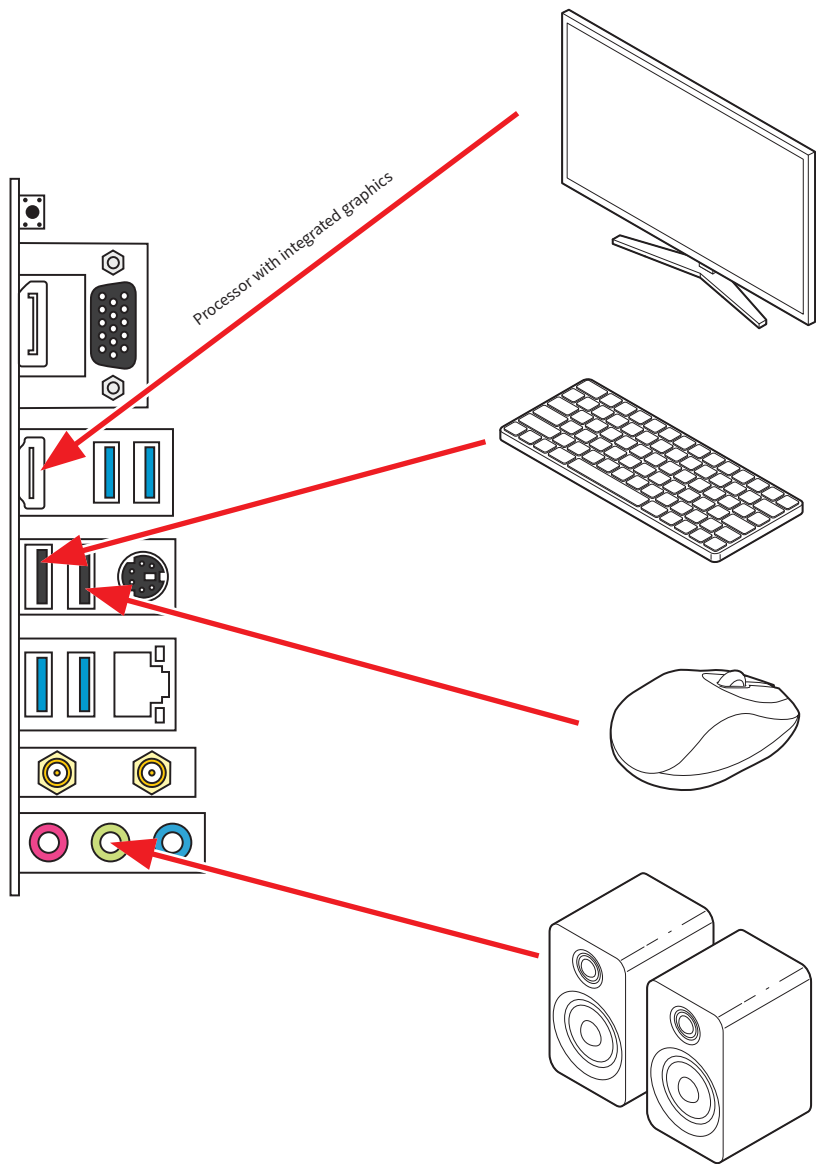
Installing a Graphics Card



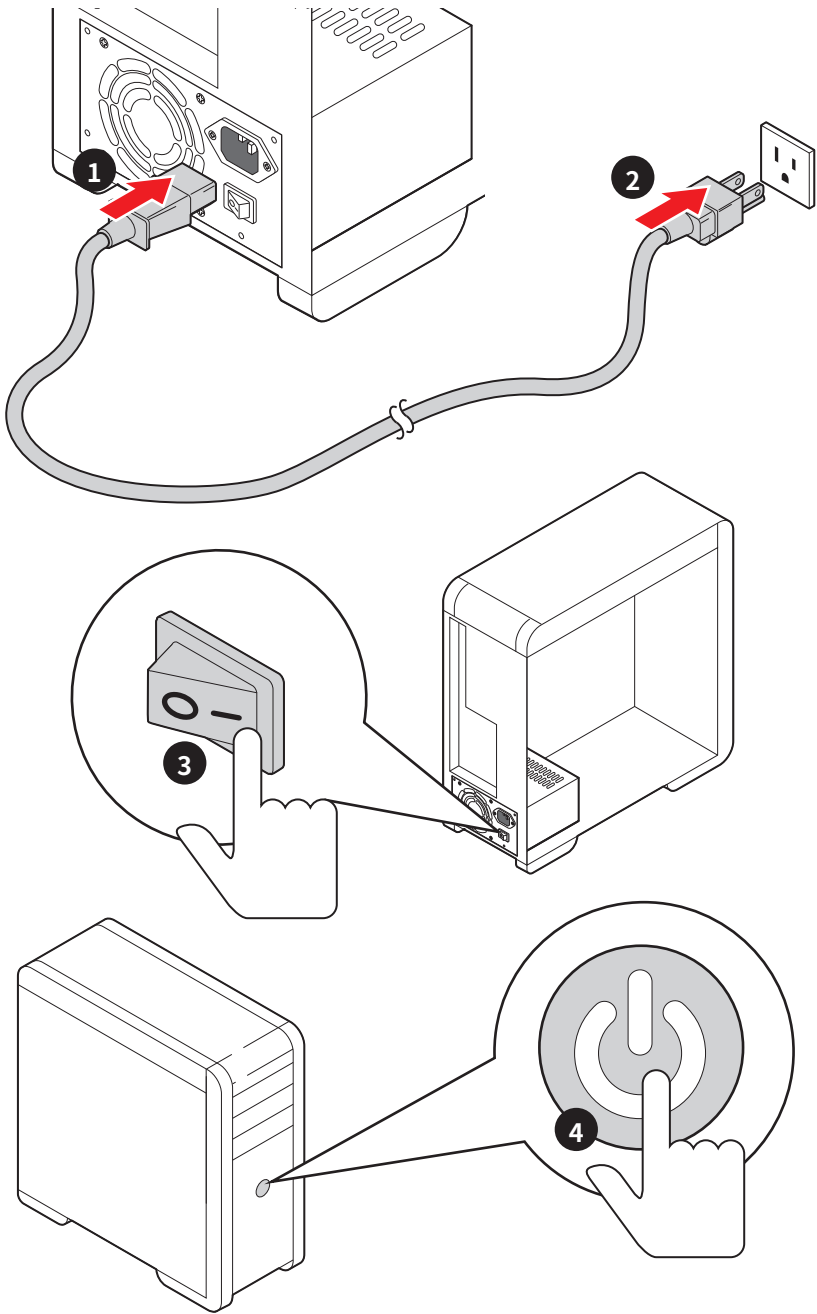
http://youtu.be/mG0GZpr9w_A



Connecting Peripheral Devices



Power On



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Connecting the Power Connectors	8
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Specifications

CPU	Supports AM4 socket 3rd Gen AMD Ryzen™ processors, and future AMD Ryzen™ processors with BIOS update
Chipset	AMD B550 Chipset
Memory	• 4x DDR4 memory slots, support up to 128GB* • Supports DDR4 1866/ 2133/ 2400/ 2667/ 2800/ 2933/ 3000/ 3066/ 3200 MHz by JEDEC • Supports DDR4 2667/ 2800 /2933 /3000 /3066 /3200 /3466 /3600/ 3733 /3866 /4000 /4133 /4266 /4400+ MHz by A-XMP OC MODE • 1DPC 1R max speed 4400 MHZ • 1DPC 2R max speed 3733 MHZ • 2DPC 1R max speed 3866 MHZ • 2DPC 2R max speed 3466 MHZ • Dual channel memory architecture • Supports non-ECC UDIMM memory • Supports ECC UDIMM memory • Supports un-buffered memory * Please refer www.msi.com for more information on compatible memory.
Expansion Slot	• 1x PCIe 3.0/ 4.0 x16 slot (PCI_E1)* • 2x PCIe 3.0 x1 slots * The supported specification depends on installed processor.
Onboard Graphics	• 1x VGA port, supports resolution of 2048x1536 @50Hz, 2048x1280 @60Hz, 1920x1200 @60Hz* • 1x DisplayPort, supports a maximum resolution of 4096x2160 @60Hz* • 1x HDMI 2.1 port, supports a maximum resolution of 4096x2160 @60Hz** • Maximum shared memory of 16 GB * Available for the processor with integrated graphics. ** Graphics specifications may vary depending on the CPU installed.

Continued on next page

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Storage	AMD B550 Chipset • 4x SATA 6Gb/s ports • 2x M.2 slots (Key M) • M2_1 slot (from AMD processor) • Supports PCIe 4.0/ 3.0 x4 * • Supports SATA 6Gb/s • Supports 2242/ 2260/ 2280 storage devices • M2_2 slot (from AMD B550 chipset) • Supports PCIe 3.0x4 • Supports 2242/ 2260/ 2280 storage devices * The supported specification depends on installed processor.
RAID	• Supports RAID 0, RAID 1 and RAID 10 for SATA storage devices • Supports RAID 0 and RAID 1 for M.2 NVMe storage devices
USB	AMD B550 Chipset • 3x USB 3.2 Gen 1 5Gbps ports (1 Type-C internal connector, and 2 ports are available through the internal USB 3.2 Gen 1 5Gbps connector) • 6x USB 2.0 ports (2 Type-A ports on the back panel, 4 ports through the internal USB 2.0 connectors) AMD Processor • 4x USB 3.2 Gen 1 5Gbps Type-A ports on the back panel
Audio	Realtek® ALC892/ ALC897 Codec • 7.1-Channel High Definition Audio
LAN	• 1x 1Gbps LAN controller

Continued on next page

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Wireless LAN & Bluetooth®	The Wireless module is pre-installed in the M.2 (Key-E) slot Wi-Fi 5 (For B550M PRO-VDH WIFI) • Supports Wi-Fi 802.11 a/b/g/n/ac, dual band (2.4GHz, 5GHz) up to 433 Mbps speed. • Supports Dual Mode Bluetooth® 2.1, 2.1+EDR, 3.0, 4.0, BLE, 4.2 Wi-Fi 6 (For B550M PRO-VDH WIFI6) • Supports Wi-Fi 802.11 a/b/g/n/ac/ax, MU-MIMO Rx, 2.4GHz-5GHz (160MHz) up to 2.4Gbps • Supports Bluetooth® 5 Wi-Fi 6E (For B550M PRO-VDH WIFI) • Supports MU-MIMO TX/RX, 2.4GHz/ 5GHz/ 6GHz* • Supports 802.11 a/ b/ g/ n/ ac/ ax • Supports Bluetooth® 5.3**, FIPS, FISMA * The usage of Wi-Fi 6GHz band relies on Windows 11 support and may depend on every country' s regulations. ** The Bluetooth version may be updated, please refer to the Wi-Fi chipset vendor' s website for details.
Internal Connectors	• 1x 24-pin ATX main power connector • 1x 8-pin ATX 12V power connector • 4x SATA 6Gb/s connectors • 2x M.2 slots (M-Key) • 1x USB 3.2 Gen 1 5Gbps Type-C port • 1x USB 3.2 Gen 1 5Gbps connector (supports additional 2 USB 3.2 Gen 1 5Gbps ports) • 2x USB 2.0 connectors (supports additional 4 USB 2.0 ports) • 1x 4-pin CPU fan connector • 1x 4-pin water-pump fan connector • 3x 4-pin system fan connectors • 1x Front panel audio connector • 2x System panel connectors • 1x Serial port connector • 1x Chassis Intrusion connector • 2x 4-pin RGB LED connectors • 2x 3-pin RAINBOW LED connectors • 1xTPM module connector • 1x Clear CMOS jumper

Continued on next page

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LED Features	• 1x EZ LED Control switch • 4x EZ Debug LED
Back Panel Connectors	• 1x Flash BIOS Button • 1x VGA port • 1x Display port • 1x HDMI port • 4x USB 3.2 Gen 1 5Gbps Type-A ports • 1x PS/2 keyboard/ mouse combo port • 2x USB 2.0 Type-A ports • 1x LAN (RJ45) port • 2x Wi-Fi Antenna connectors • 3x Audio jacks
I/O Controller	NUVOTON NCT6687-R Controller Chip
Hardware Monitor	• CPU/ System/ Chipset temperature detection • CPU/ System/ Pump fan speed detection • CPU/ System/ Pump fan speed control
Form Factor	• Micro-ATX Form Factor • 9.6 in. x 9.6 in. (24.4 cm x 24.4 cm)
BIOS Features	• 1x 256 Mb flash • UEFI AMI BIOS • ACPI 6.0, SMBIOS 2.8 • Multi-language
Software	• Drivers • DRAGON CENTER • CPU-Z MSI GAMING • Google Chrome™, Google Toolbar, Google Drive • Norton™ Internet Security Solution

Continued on next page

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Dragon Center Features	• Mystic Light • User Scenario • Hardware Monitor • True Color • Live Update • Speed Up • Smart Tool • Super Charger  Please refer to http://download.msi.com/manual/mb/DRAGONCENTER2.pdf for more details.
Special Features	• Audio • Audio Boost • Network • Intel WiFi • Cooling • Pump Fan • Smart Fan Control • LED • Mystic Light Extension (RAINBOW/RGB) • EZ LED Control • EZ DEBUG LED • Performance • DDR4 Boost • Core Boost • Creator Genie • Front USB Type-C • Protection • PCI-E Steel Armor • Experience • Dragon Center • Click BIOS 5 • Flash BIOS Button

Package contents

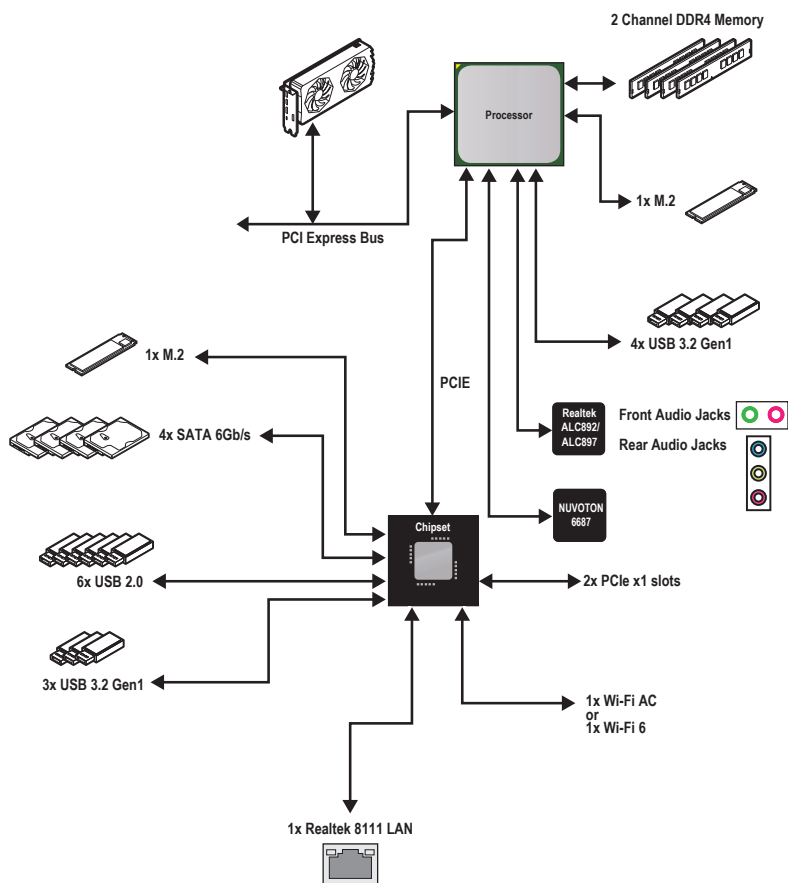
Please check the contents of your motherboard package. It should contain:

Motherboard	B550M PRO-VDH WIFI/ B550M PRO-VDH WIFI6	
Cable	SATA 6G cables (2 cables/pack)	1
Accessories	Wi-Fi antenna	1
	M.2 screws (3 pcs./ pack)	1
Documentation	Quick installation guide	1
	MSI components compatibility & reward program card	1

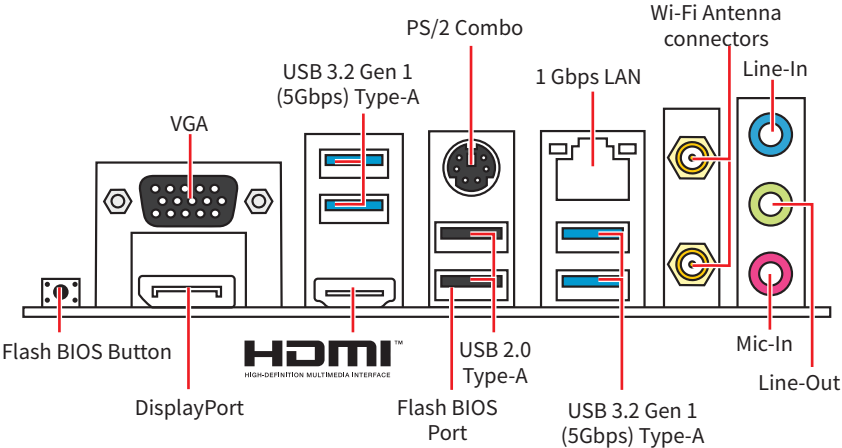


If any of the above items are damaged or missing, please contact your retailer.

Block Diagram

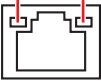


Rear I/O Panel



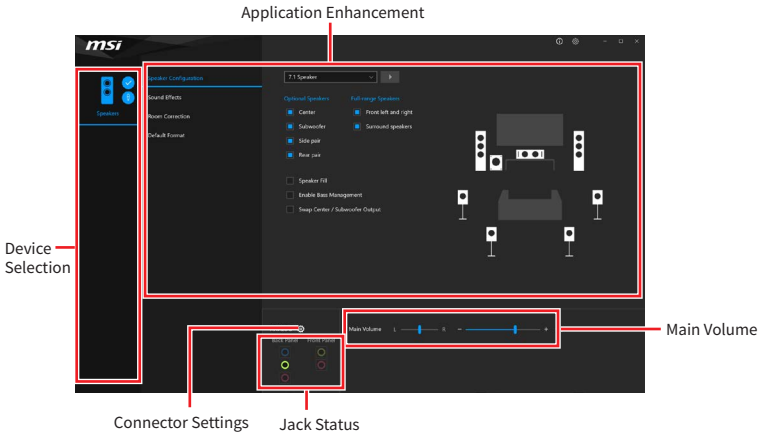
- **Flash BIOS Port/ Button** - Please refer to page 48 for Updating BIOS with Flash BIOS Button.

LAN Port LED Status Table

Link/ Activity LED			Speed LED	
Status	Description		Status	Description
Off	No link		Off	10 Mbps connection
Yellow	Linked		Green	100 Mbps connection
Blinking	Data activity		Orange	1 Gbps connection

Realtek Audio Console

After Realtek Audio Console is installed. You can use it to change sound settings to get better sound experience.

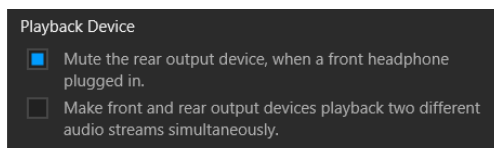


- **Device Selection** - allows you to select a audio output source to change the related options. The **check** sign indicates the devices as default.
- **Application Enhancement** - the array of options will provide you a complete guidance of anticipated sound effect for both output and input device.
- **Main Volume** - controls the volume or balance the right/left side of the speakers that you plugged in front or rear panel by adjust the bar.
- **Jack Status** - depicts all render and capture devices currently connected with your computer.
- **Connector Settings** - configures the connection settings.

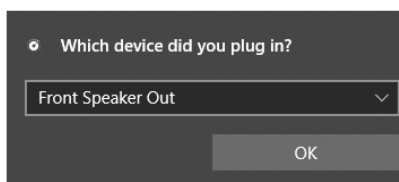
Audio 7.1-channel Configuration

To configure 7.1-channel audio, you have to connect front audio I/O module to JAUD1 connector and follow the below steps.

1. Click on the Realtek Audio Console > Device advanced settings to open the dialog below.



2. Select **Mute the rear output device, when a front headphone plugged in.**
3. Plug your speakers to audio jacks on rear and front I/O panel. When you plug a device into an audio jack, a dialogue window will pop up asking you which device is current connected.

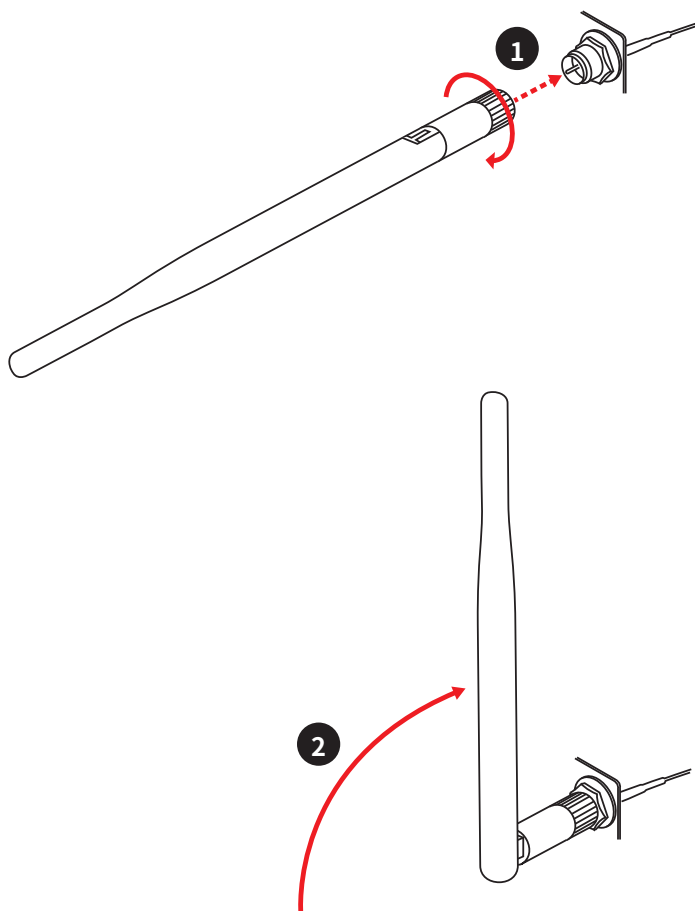


Important

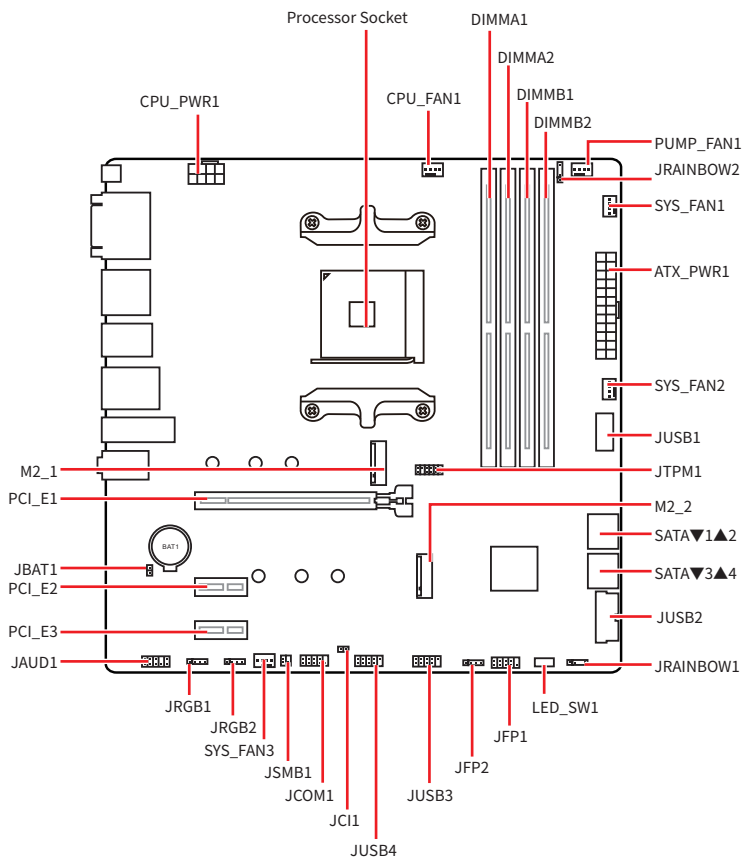
The pictures above for reference only and may vary from the product you purchased.

Installing antennas

1. Screw the antennas tight to the antenna connectors as shown below.
2. Orient the antennas.



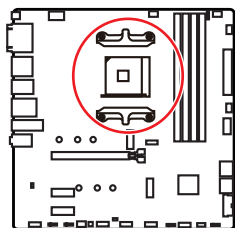
Overview of Components



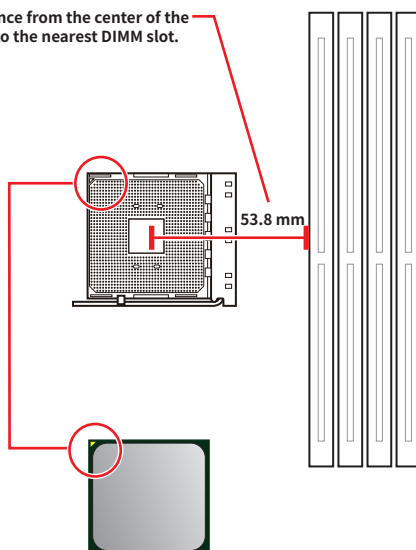
Component Contents

Port Name	Port Type	Page
CPU_FAN1, PUMP_FAN1, SYS_FAN1~3	Fan Connectors	36
CPU_PWR1, ATX_PWR1	Power Connectors	33
DIMMA1, DIMMA2, DIMMB1, DIMMB2	DIMM Slots	28
JAUD1	Front Audio Connector	32
JBAT1	Clear CMOS (Reset BIOS) Jumper	38
JCI1	Chassis Intrusion Connector	37
JCOM1	Serial Port Connector	38
JFP1, JFP2	Front Panel Connectors	32
JRAINBOW1~2	Addressable RGB LED connectors	40
JRGB1~2	RGB LED connector	39
JTPM1	TPM Module Connector	35
JUSB1	USB 3.2 Gen 1 5Gbps Type-C Connector	34
JUSB2	USB 3.2 Gen 1 5Gbps Connector	34
JUSB3~4	USB 2.0 Connectors	35
LED_SW1	EZ LED Control	41
M2_1~2	M.2 Slots (Key M)	30
PCI_E1~3	PCIe Expansion Slots	29
Processor Socket	Socket AM4	27
SATA1~4	SATA 6Gb/s Connectors	29

Processor Socket



Distance from the center of the CPU to the nearest DIMM slot.



Introduction to the AM4 CPU

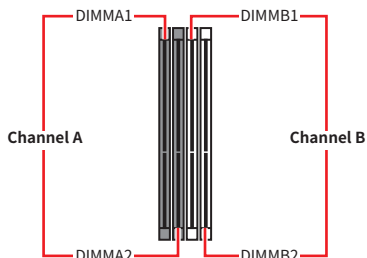
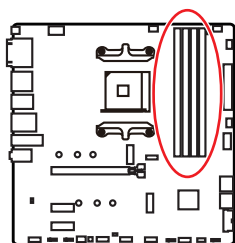
The surface of the AM4 CPU has a yellow triangle to assist in correctly lining up the CPU for motherboard placement. The yellow triangle is the Pin 1 indicator.



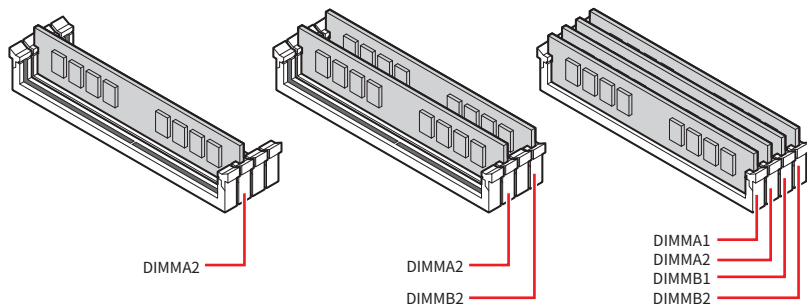
Important

- When changing the processor, the system configuration could be cleared and reset BIOS to default values, due to the AM4 processor's architecture.
- Always unplug the power cord from the power outlet before installing or removing the CPU.
- When installing a CPU, always remember to install a CPU heatsink. A CPU heatsink is necessary to prevent overheating and maintain system stability.
- Confirm that the CPU heatsink has formed a tight seal with the CPU before booting your system.
- Overheating can seriously damage the CPU and motherboard. Always make sure the cooling fans work properly to protect the CPU from overheating. Be sure to apply an even layer of thermal paste (or thermal tape) between the CPU and the heatsink to enhance heat dissipation.
- If you purchased a separate CPU and heatsink/ cooler, Please refer to the documentation in the heatsink/ cooler package for more details about installation.
- This motherboard is designed to support overclocking. Before attempting to overclock, please make sure that all other system components can tolerate overclocking. Any attempt to operate beyond product specifications is not recommended. MSI® does not guarantee the damages or risks caused by inadequate operation beyond product specifications.

DIMM Slots



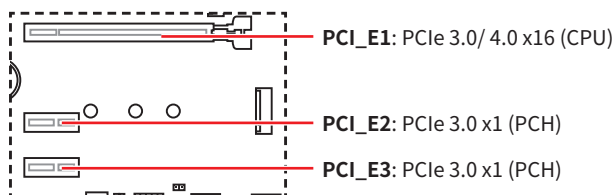
Memory module installation recommendation



Important

- Always insert memory modules in the **DIMMA2** slot first.
- Due to chipset resource usage, the available capacity of memory will be a little less than the amount of installed.
- Based on CPU specification, the Memory DIMM voltage below 1.35V is suggested to protect the CPU.
- To ensure system stability for Dual channel mode, memory modules must be of the same type, number and density.
- Some memory modules may operate at a lower frequency than the marked value when overclocking due to the memory frequency operates dependent on its Serial Presence Detect (SPD). Go to BIOS and find the **DRAM Frequency** to set the memory frequency if you want to operate the memory at the marked or at a higher frequency.
- It is recommended to use a more efficient memory cooling system for full DIMMs installation or overclocking.
- The stability and compatibility of installed memory module depend on installed CPU and devices when overclocking.
- Please refer www.msi.com for more information on compatible memory.

PCI_E1~3: PCIe Expansion Slots

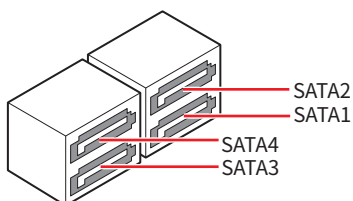
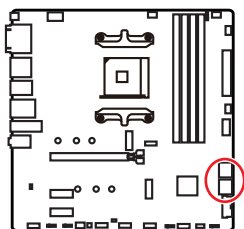


Important

- If you install a large and heavy graphics card, you need to use a tool such as **MSI Gaming Series Graphics Card Bolster** to support its weight to prevent deformation of the slot.
- When adding or removing expansion cards, always turn off the power supply and unplug the power supply power cable from the power outlet. Read the expansion card's documentation to check for any necessary additional hardware or software changes.

SATA1~4: SATA 6Gb/s Connectors

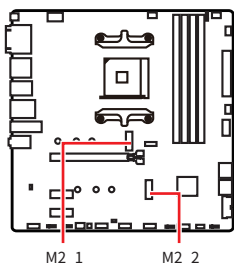
These connectors are SATA 6Gb/s interface ports. Each connector can connect to one SATA device.



Important

- Please do not fold the SATA cable at a 90-degree angle. Data loss may result during transmission otherwise.
- SATA cables have identical plugs on either sides of the cable. However, it is recommended that the flat connector be connected to the motherboard for space saving purposes.

M2_1~2: M.2 Slots (Key M)



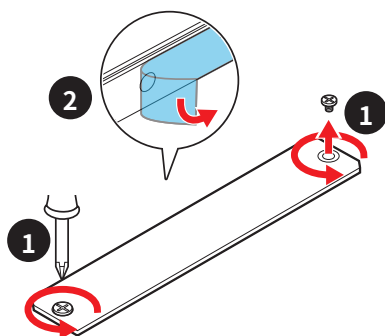
Video Demonstration

Watch the video to learn how to Install M.2 module.

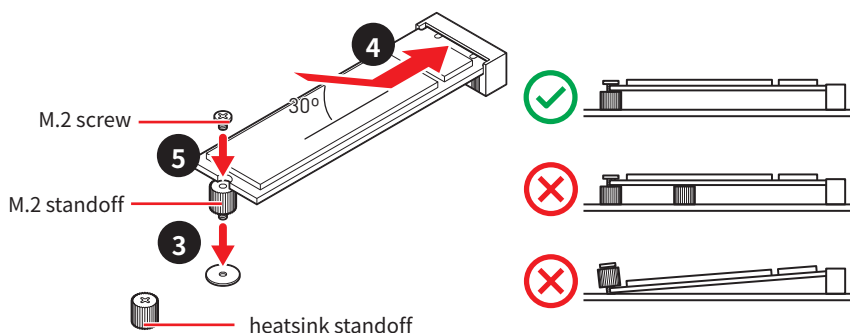
<http://youtu.be/JCTFABytrYA>

M2_1 slot installation

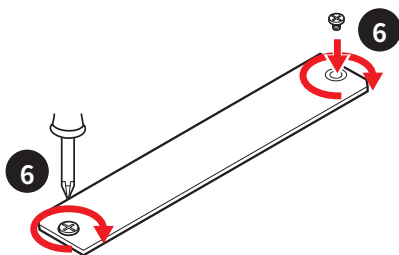
1. Loosen the screws of M.2 SHIELD FROZR heatsink.
2. Remove the M.2 SHIELD FROZR and remove the protective films from the thermal pads.



3. Move and fasten the M.2 standoff to the appropriate position for your M.2 SSD, or remove the M.2 standoff if your M.2 SSD length is same as the length of M.2 heatsink to avoid damage to the M.2 SSD.
4. Insert your M.2 SSD into the M.2 slot at a 30-degree angle.
5. Secure the M.2 SSD in place with the M.2 screw, or skip this step if you remove the M.2 standoff in step 3.



6. Put the M.2 SHIELD FROZR heatsink back in place and secure it.

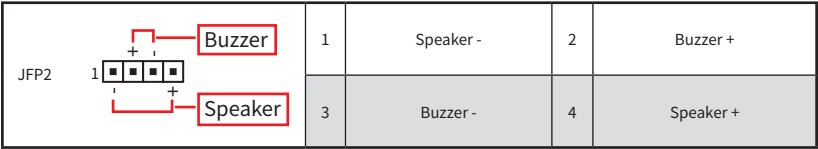
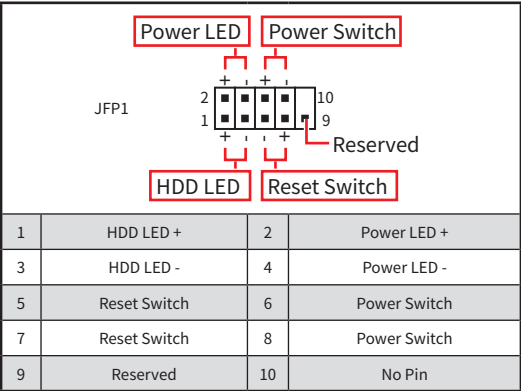
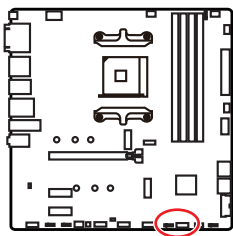


M2_2 slot installation

1. Move the position of the standoffs according to your M.2 SSDs length if need.
2. Insert your M.2 SSD into the M.2 slot at a 30-degree angle.
3. Secure the M.2 SSD in place with the supplied M.2 screw.

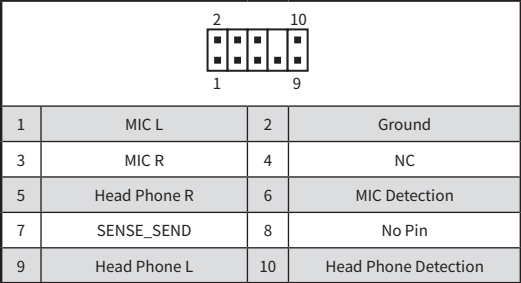
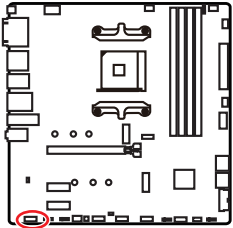
JFP1, JFP2: Front Panel Connectors

These connectors connect to the switches and LEDs on the front panel.



JAUD1: Front Audio Connector

This connector allows you to connect audio jacks on the front panel.



These connectors allow you to connect an ATX power supply.

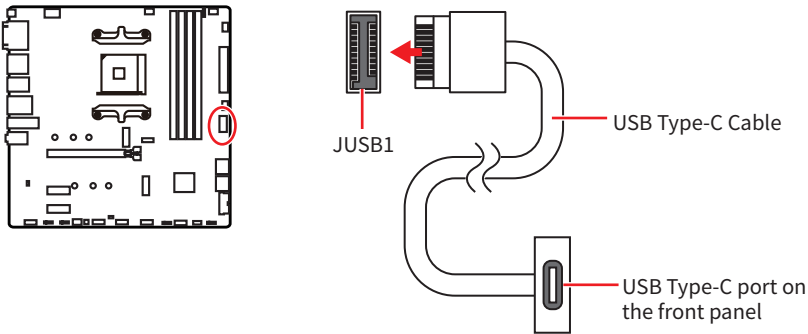
Pin	Signal	Pin	Signal
1	+3.3V	13	+3.3V
2	+3.3V	14	-12V
3	Ground	15	Ground
4	+5V	16	PS-ON#
5	Ground	17	Ground
6	+5V	18	Ground
7	Ground	19	Ground
8	PWR OK	20	Res
9	5VSB	21	+5V
10	+12V	22	+5V
11	+12V	23	+5V
12	+3.3V	24	Ground



Overview of Components 33

JUSB1: USB 3.2 Gen 1 5Gbps Type-C Connector

This connector allows you to connect USB 3.2 Gen 1 5Gbps Type-C connector on the front panel. The connector possesses a foolproof design. When you connect the cable, be sure to connect it with the corresponding orientation.



JUSB2: USB 3.2 Gen 1 5Gbps Connector

This connector allows you to connect USB 3.2 Gen 1 5Gbps ports on the front panel.

The diagram shows the JUSB2 connector on the motherboard, which is a 20-pin USB connector. A front panel port is shown connected to the JUSB2 connector. The front panel port is labeled "JUSB2" and the motherboard connector is labeled "JUSB2".

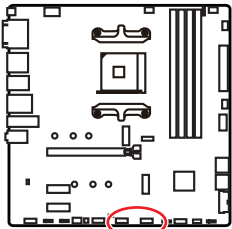
10	11		
1	20		
1	Power	11	USB2.0+
2	USB3_RX_DN	12	USB2.0-
3	USB3_RX_DP	13	Ground
4	Ground	14	USB3_TX_C_DP
5	USB3_TX_C_DN	15	USB3_TX_C_DN
6	USB3_TX_C_DP	16	Ground
7	Ground	17	USB3_RX_DP
8	USB2.0-	18	USB3_RX_DN
9	USB2.0+	19	Power
10	Ground	20	No Pin

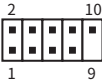
Important

Note that the Power and Ground pins must be connected correctly to avoid possible damage.

JUSB3~4: USB 2.0 Connectors

These connectors allow you to connect USB 2.0 ports on the front panel.



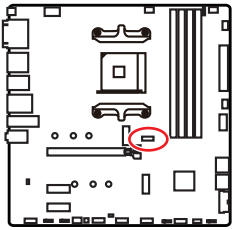
			
1	VCC	2	VCC
3	USB0-	4	USB1-
5	USB0+	6	USB1+
7	Ground	8	Ground
9	No Pin	10	NC

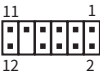
Important

- Note that the VCC and Ground pins must be connected correctly to avoid possible damage.
- In order to recharge your iPad,iPhone and iPod through USB ports, please install MSI® DRAGON CENTER utility.

JTPM1: TPM Module Connector

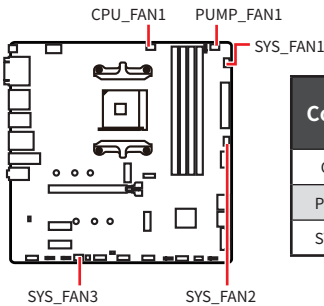
This connector is for TPM (Trusted Platform Module). Please refer to the TPM security platform manual for more details and usages.



			
1	SPI Power	2	SPI Chip Select
3	Master In Slave Out (SPI Data)	4	Master In Slave In (SPI Data)
5	Reserved	6	SPI Clock
7	Ground	8	SPI Reset
9	Reserved	10	No Pin
11	Reserved	12	Interrupt Request

CPU_FAN1, PUMP_FAN1, SYS_FAN1~3: Fan Connectors

Fan connectors can be classified as PWM (Pulse Width Modulation) Mode or DC Mode. PWM Mode fan connectors provide constant 12V output and adjust fan speed with speed control signal. DC Mode fan connectors control fan speed by changing voltage. The auto mode fan connectors can automatically detect PWM and DC mode. However, you can follow the instruction below to adjust the fan connector to PWM or DC Mode manually.

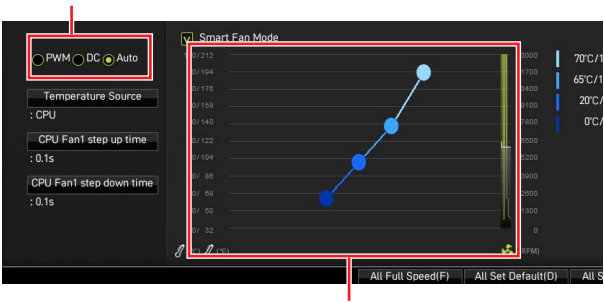


Connector	Default fan mode	Max. current	Max. power
CPU_FAN1	Auto mode	2A	24W
PUMP_FAN1	PWM mode	3A	36W
SYS_FAN1~3	DC mode	1A	12W

Switching fan mode and adjusting fan speed

You can switch between PWM mode and DC mode and adjust fan speed in BIOS > **HARDWARE MONITOR**.

Select **PWM** mode or **DC** mode



There are gradient points of the fan speed that allow you to adjust fan speed in relation to CPU temperature.

Important

Make sure fans are working properly after switching the PWM/ DC mode.

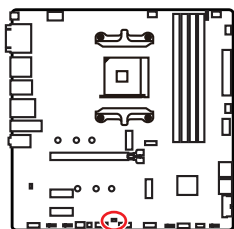
Pin definition of fan connectors

1  PWM Mode pin definition			
1	Ground	2	+12V
3	Sense	4	Speed Control Signal

1  DC Mode pin definition			
1	Ground	2	Voltage Control
3	Sense	4	NC

JCI1: Chassis Intrusion Connector

This connector allows you to connect the chassis intrusion switch cable.



Normal
(default)



Trigger the chassis
intrusion event

Using chassis intrusion detector

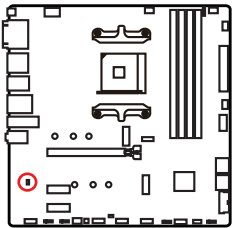
1. Connect the **JCI1** connector to the chassis intrusion switch/ sensor on the chassis.
2. Close the chassis cover.
3. Go to **BIOS > SETTINGS > Security > Chassis Intrusion Configuration**.
4. Set **Chassis Intrusion** to **Enabled**.
5. Press **F10** to save and exit and then press the **Enter** key to select **Yes**.
6. Once the chassis cover is opened again, a warning message will be displayed on screen when the computer is turned on.

Resetting the chassis intrusion warning

1. Go to **BIOS > SETTINGS > Security > Chassis Intrusion Configuration**.
2. Set **Chassis Intrusion** to **Reset**.
3. Press **F10** to save and exit and then press the **Enter** key to select **Yes**.

JBAT1: Clear CMOS (Reset BIOS) Jumper

There is CMOS memory onboard that is external powered from a battery located on the motherboard to save system configuration data. If you want to clear the system configuration, set the jumpers to clear the CMOS memory.



Keep Data
(default)



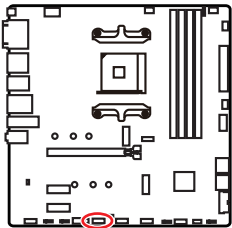
Clear CMOS/
Reset BIOS

Resetting BIOS to default values

- 1. Power off the computer and unplug the power cord.
- 2. Use a jumper cap to short **JBAT1** for about 5-10 seconds.
- 3. Remove the jumper cap from **JBAT1**.
- 4. Plug the power cord and Power on the computer.

JCOM1: Serial Port Connector

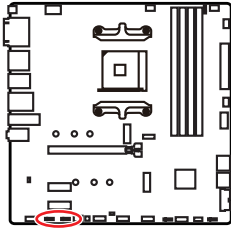
This connector allows you to connect the optional serial port with bracket.



210 19			
1	DCD	2	SIN
3	SOUT	4	DTR
5	Ground	6	DSR
7	RTS	8	CTS
9	RI	10	No Pin

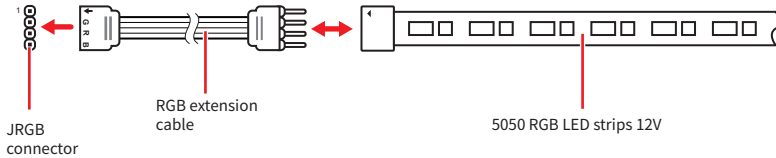
JRGB1~2: RGB LED connector

The JRGB connectors allow you to connect the 5050 RGB LED strips 12V.

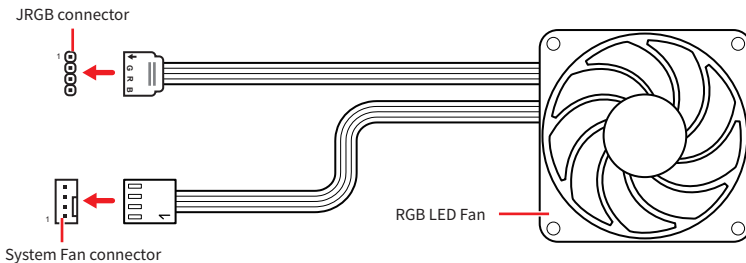


1			
1	+12V	2	G
3	R	4	B

RGB LED Strip Connection



RGB LED Fan Connection

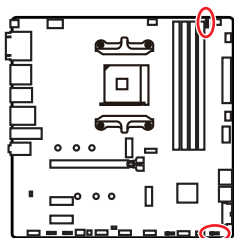


Important

- The JRGB connector supports up to 2 meters continuous 5050 RGB LED strips (12V/G/R/B) with the maximum power rating of 3A (12V).
- Always turn off the power supply and unplug the power cord from the power outlet before installing or removing the RGB LED strip.
- Please use MSI' s software to control the extended LED strip.

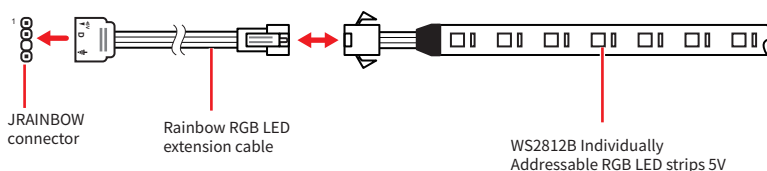
JRAINBOW1~2: Addressable RGB LED connectors

The JRAINBOW connectors allow you to connect the WS2812B Individually Addressable RGB LED strips 5V.

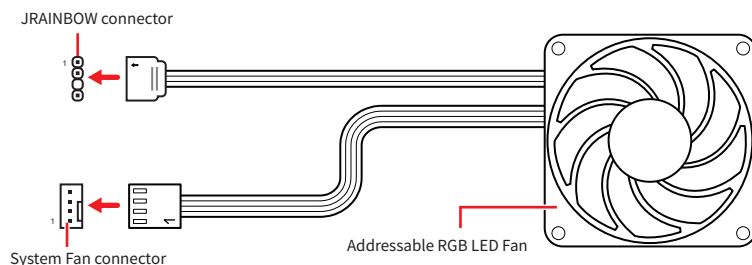


 JRAINBOW1		 JRAINBOW2	
1	+5V	2	Data
3	No Pin	4	Ground

Addressable RGB LED Strip Connection



Addressable RGB LED Fan Connection



CAUTION

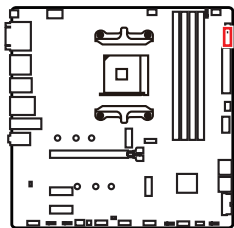
Do not connect the wrong type of LED strips. The JRGB connector and the JRAINBOW connector provide different voltages, and connecting the 5V LED strip to the JRGB connector will result in damage to the LED strip.

Important

- The JRAINBOW connector supports up to 75 LEDs WS2812B Individually Addressable RGB LED strips (5V/Data/Ground) with the maximum power rating of 3A (5V). In the case of 20% brightness, the connector supports up to 200 LEDs.
- Always turn off the power supply and unplug the power cord from the power outlet before installing or removing the RGB LED strip.
- Please use MSI's software to control the extended LED strip.

EZ Debug LED

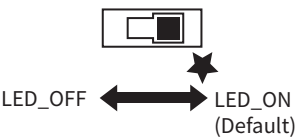
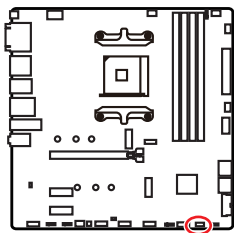
These LEDs indicate the debug status of the motherboard.



- ☐ **CPU** - indicates CPU is not detected or fail.
- ☐ **DRAM** - indicates DRAM is not detected or fail.
- ☐ **VGA** - indicates GPU/ PCIE/ M.2 device is not detected or fail.
- ☐ **BOOT** - indicates the booting device is not detected or fail.

LED_SW1: EZ LED Control

This switch is used to switch on/ off all the LEDs of motherboard.



Installing OS, Drivers & Utilities

Please download and update the latest utilities and drivers at www.msi.com

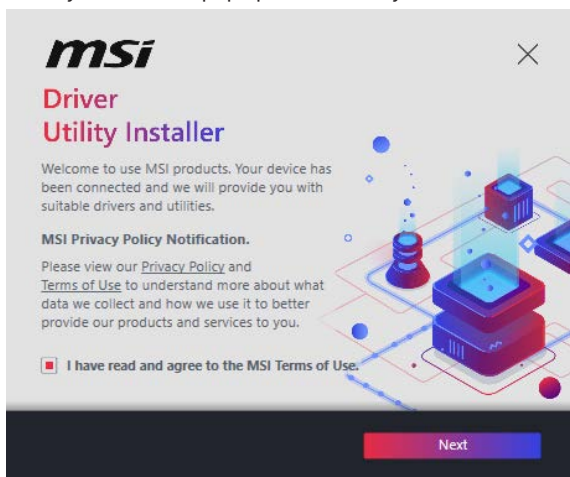
Installing Windows® 10

1. Power on the computer.
2. Insert the Windows® 10 installation disc/USB into your computer.
3. Press the **Restart** button on the computer case.
4. Press **F11** key during the computer POST (Power-On Self Test) to get into Boot Menu.
5. Select the Windows® 10 installation disc/USB from the Boot Menu.
6. Press any key when screen shows **Press any key to boot from CD or DVD...** message.
7. Follow the instructions on the screen to install Windows® 10.

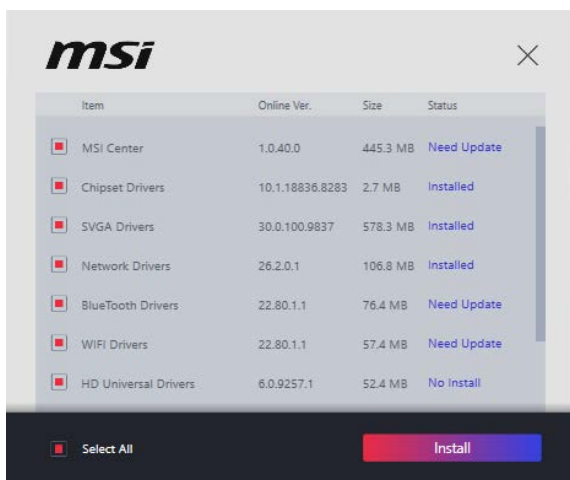
Installing Drivers with MSI Driver Utility Installer

Important

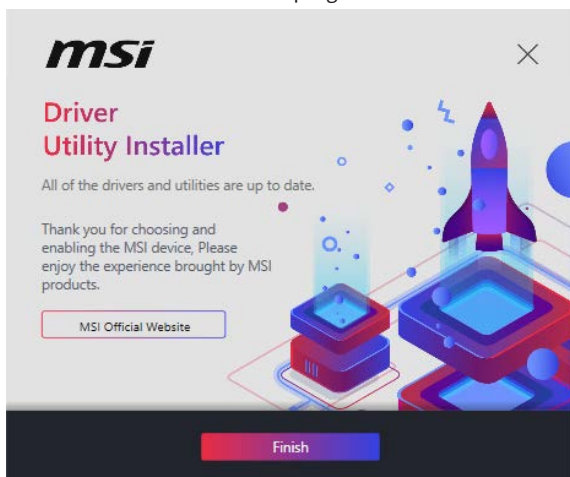
- Some new network chips have not been natively supported by Windows 10/ Windows 11. It is recommended that the **LAN driver** be installed before installing drivers with MSI Driver Utility Installer. Please refer to www.msi.com to install the LAN driver for your motherboard.
 - The MSI Driver Utility Installer will only pop up once. If you cancel or close it during the process, please refer to the Live Update chapter of the MSI Center manual to install the drivers. You can also go to www.msi.com to search your motherboard and download the drivers.
 - MSI Driver Utility Installer needs to be installed over the internet.
1. Start up your computer in Windows 10/ Windows 11.
 2. Select **Start > Settings > Windows Update**, and then select **Check** for updates.
 3. MSI Driver Utility Installer will pop up automatically.



4. Select the **I have read and agree to the MSI Terms of Use** check box, and then click Next.



5. Check the **Select All** checkbox in the lower-left corner and click **Install** to install MSI Center and drivers. The installation progress will be shown at the bottom.



6. Once the progress has completed, click **Finish**.

UEFI BIOS

MSI UEFI BIOS is compatible with UEFI (Unified Extensible Firmware Interface) architecture. UEFI has many new functions and advantages that traditional BIOS cannot achieve, and it will completely replace BIOS in the future. The MSI UEFI BIOS uses UEFI as the default boot mode to take full advantage of the new chipset's capabilities. However, it still has a CSM (Compatibility Support Module) mode to be compatible with older devices. That allows you to replace legacy devices with UEFI compatible devices during the transition.



Important

The term BIOS in this user guide refers to UEFI BIOS unless otherwise noted.

UEFI advantages

- Fast booting - UEFI can directly boot the operating system and save the BIOS self-test process. And also eliminates the time to switch to CSM mode during POST.
- Supports for hard drive partitions larger than 2 TB.
- Supports more than 4 primary partitions with a GUID Partition Table (GPT).
- Supports unlimited number of partitions.
- Supports full capabilities of new devices - new devices may not provide backward compatibility.
- Supports secure startup - UEFI can check the validity of the operating system to ensure that no malware tampers with the startup process.

Incompatible UEFI cases

- **32-bit Windows operating system** - this motherboard supports only Windows 10 64-bit operating system.
- **Older graphics card** - the system will detect your graphics card. When display a warning message **There is no GOP (Graphics Output protocol) support detected in this graphics card.**

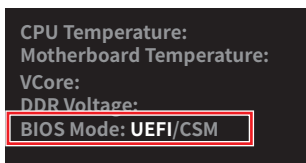


Important

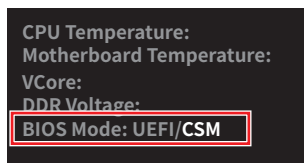
We recommend that you to use a GOP/ UEFI compatible graphics card.

How to check the BIOS mode?

After entering the BIOS, find the BIOS Mode at the top of the screen.



UEFI boot mode



CSM boot mode

BIOS Setup

The default settings offer the optimal performance for system stability in normal conditions. You should **always keep the default settings** to avoid possible system damage or failure booting unless you are familiar with BIOS.



Important

- BIOS items are continuously update for better system performance. Therefore, the description may be slightly different from the latest BIOS and should be for reference only. You could also refer to the **HELP** information panel for BIOS item description.
- The pictures in this chapter are for reference only and may vary from the product you purchased.
- The BIOS items will vary with the processor.

Entering BIOS Setup

Press **Delete** key, when the **Press DEL key to enter Setup Menu, F11 to enter Boot Menu** message appears on the screen during the boot process.

Function key

- F1:** General Help list
- F2:** Add/ Remove a favorite item
- F3:** Enter Favorites menu
- F4:** Enter CPU Specifications menu
- F5:** Enter Memory-Z menu
- F6:** Load optimized defaults
- F7:** Switch between Advanced mode and EZ mode
- F8:** Load Overclocking Profile
- F9:** Save Overclocking Profile
- F10:** Save Change and Reset*
- F12:** Take a screenshot and save it to USB flash drive (FAT/ FAT32 format only).
- Ctrl+F:** Enter Search page

* When you press F10, a confirmation window appears and it provides the modification information. Select between Yes or No to confirm your choice.

Resetting BIOS

You might need to restore the default BIOS setting to solve certain problems. There are several ways to reset BIOS:

- Go to BIOS and press **F6** to load optimized defaults.
- Short the **Clear CMOS** jumper on the motherboard.



Important

*Be sure the computer is off before clearing CMOS data. Please refer to the **Clear CMOS** jumper section for resetting BIOS.*

Updating BIOS

Updating BIOS with M-FLASH

Before updating:

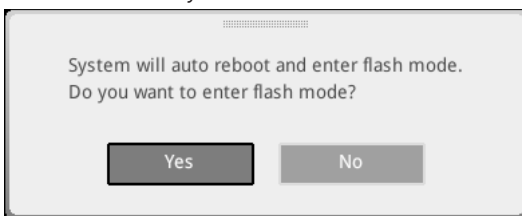
Please download the latest BIOS file that matches your motherboard model from MSI website. And then save the BIOS file into the USB flash drive.

Updating BIOS:

1. Insert the USB flash drive that contains the update file into the USB port.
2. Please refer the following methods to enter flash mode.
 - Reboot and press **Ctrl + F5** key during POST and click on **Yes** to reboot the system.

Press <Ctrl+F5> to activate M-Flash for BIOS update.

- Reboot and press **Del** key during POST to enter BIOS. Click the **M-FLASH** button and click on **Yes** to reboot the system.



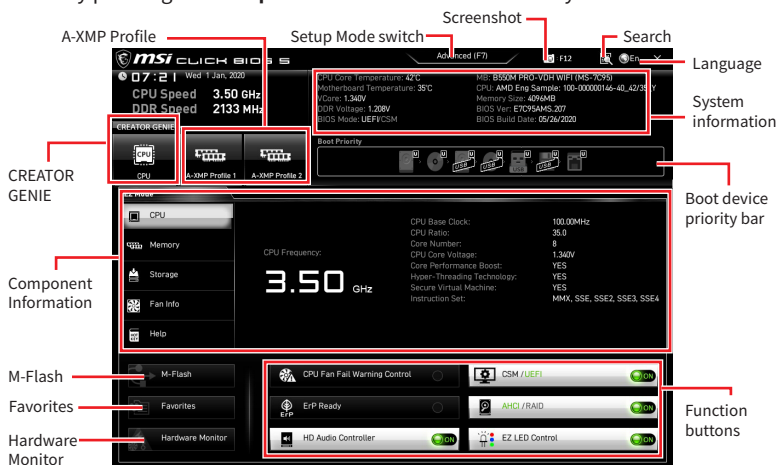
3. Select a BIOS file to perform the BIOS update process.
4. When prompted click on **Yes** to start recovering BIOS.
5. After the flashing process is 100% completed, the system will reboot automatically.

Updating BIOS with Flash BIOS Button

1. Please download the latest BIOS file that matches your motherboard model from the MSI® website.
2. Rename the BIOS file to **MSI.ROM**, and save it to the root of your USB flash drive.
3. Connect the power supply to **CPU_PWR1** and **ATX_PWR1**. (No need to install CPU and memory.)
4. Plug the USB flash drive that contains the MSI.ROM file into the **Flash BIOS Port** on the rear I/O panel.
5. Press the **Flash BIOS** Button to flash BIOS, and the LED starts flashing.
6. The LED will be turned off when the process is completed.

EZ Mode

At EZ mode, it provides the basic system information and allows you to configure the basic setting. To configure the advanced BIOS settings, please enter the Advanced Mode by pressing the **Setup Mode switch** or **F7** function key.



- **CREATOR GENIE** - click on it to toggle the CREATOR GENIE for performance optimization. This function is only available when both of the motherboard and CPU are supporting this function.



Important

Please don't make any changes in OC menu and don't load defaults to keep the optimal performance and system stability after activating the CREATOR GENIE function.

- **A-XMP Profile** - allows you to select the A-XMP profile for memory to overclock. This function is only available when the system, memory and CPU are supporting this function.
- **Setup Mode switch** - press this tab or the **F7** key to switch between Advanced mode and EZ mode.
- **Screenshot** - click on this tab or the **F12** key to take a screenshot and save it to USB flash drive (FAT/ FAT32 format only).
- **Search** - click on this tab or the **Ctrl+F** keys to enter the search page. It allows you to search by BIOS item name. Move the mouse over a blank space and right click the mouse to exit the search page.



Important

In search page, only the F6, F10 and F12 function keys are available.

- **Language** - allows you to select language of BIOS setup.

- **System information** - shows the CPU/ DDR speed, CPU/ MB temperature, MB/ CPU type, memory size, CPU/ DDR voltage, BIOS version and build date.
- **Boot device priority bar** - you can move the device icons to change the boot priority. The boot priority from high to low is left to right.
- **Component Information** - click on the **CPU**, **Memory**, **Storage**, **Fan Info** and **Help** buttons to show the information of connected component.
- **Function buttons** - enable or disable these functions by clicking on these buttons. The function is enabled when the button shows **ON**.

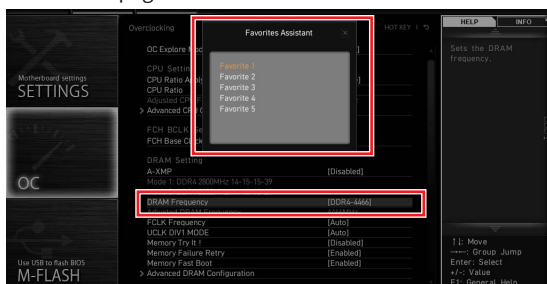
Important

The function buttons will vary with the motherboard you purchased.

- **M-Flash** - click on this button to enter the **M-Flash** menu that provides the way to update BIOS with a USB flash drive.
- **Hardware Monitor** - click on this button to enter the **Hardware Monitor** menu that allows you to manually control the fan speed by percentage.
- **Favorites** - click on this button or press the **F3** key to show the **Favorites** window. It provides 5 menus for you to create personal BIOS menu where you can save and access favorite/ frequently-used BIOS setting items.



- **To add a BIOS item to a favorite menu**
 1. Select a BIOS item not only on BIOS menu but also on search page.
 2. Right-click or press **F2** key.
 3. Choose a favorite page and click on **OK**.

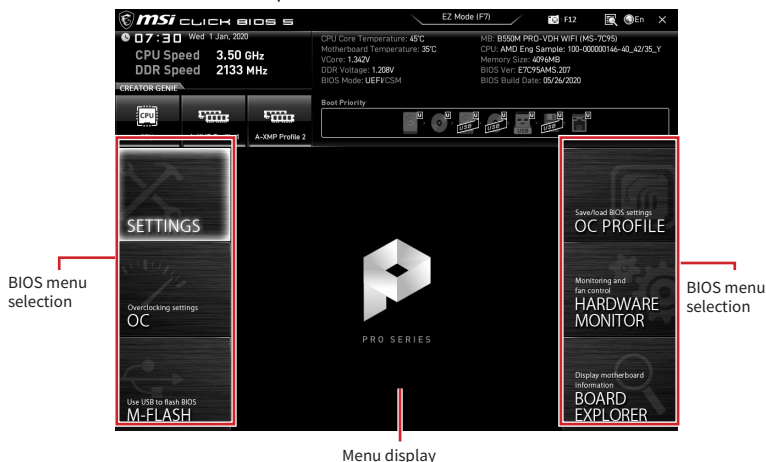


- **To delete a BIOS item from favorite menu**
 1. Select a BIOS item on favorite menu.
 2. Right-click or press **F2** key.
 3. Choose **Delete** and click on **OK**.



Advanced Mode

Press **Setup Mode switch** or **F7** function key can switch between EZ Mode and Advanced Mode in BIOS setup.



- **BIOS menu selection** - the following options are available:
 - **SETTINGS** - allows you to specify the parameters for chipset and boot devices.
 - **OC** - allows you to adjust the frequency and voltage. Increasing the frequency may get better performance.
 - **M-FLASH** - provides the way to update BIOS with a USB flash drive.
 - **OC PROFILE** - allows you to manage overclocking profiles.
 - **HARDWARE MONITOR** - allows you to set the speeds of fans and monitor voltages of system.
 - **BOARD EXPLORER** - provides the information of installed devices on this motherboard.
- **Menu display** - provides BIOS setting items and information to be configured.

SETTINGS Menu

This menu allows you to specify the parameters for system, chipset and boot devices.

▸ System Status sub-menu

The System Status sub-menu allows you to set the system clock and view system information.

▸ System Date

Sets the system date. Use tab key to switch between date elements.

The format is <day> <month> <date> <year>.

<day> Day of the week, from Sun to Sat, determined by BIOS. Read-only.

<month> The month from Jan. through Dec.

<date> The date from 1 to 31 can be keyed by numeric function keys.

<year> The year can be adjusted by users.

▸ System Time

Sets the system time. Use tab key to switch between time elements.

The time format is <hour> <minute> <second>.

▸ SATA PortX/ M2_X

Shows the information of connected SATA/ M.2 device.



Important

If the connected SATA/ M.2 device is not displayed, turn off computer and re-check SATA/ M.2 cable and power cable connections of the device and motherboard.

▸ System Information

Shows detailed system information, including CPU type, BIOS version, and Memory (read only).

▸ DMI Information

Shows system information, desktop board information and chassis information.

▸ Advanced sub-menu

The Advanced sub-menu allows you to adjust and set the parameters and behaviors of PCIe, ACPI, integrated peripherals, integrated graphics, USB, power management and Windows.

▸ PCI Subsystem Settings

Sets PCI, PCI express interface protocol and latency timer. Press **Enter** to enter the sub-menu.

▸ ACPI Settings

Sets ACPI parameters of onboard power LED behaviors.

▸ Integrated Peripherals sub-menu

Sets integrated peripherals' parameters, such as LAN, Wi-Fi, HDD, SSD, USB and audio.

► **Integrated Graphics Configuration sub-menu (optional)**

Adjusts integrated graphics settings for optimum system. This sub-menu is only available when using the CPU which integrate with IGP.

► **USB Configuration sub-menu**

Sets the onboard USB controller and device function. Press **Enter** to enter the sub-menu.

► **Power Management Setup sub-menu**

Sets system Power Management of ErP and AC Power Loss behaviors. You can optimize the system power consumption in this menu.

► **Windows OS Configuration**

Sets Windows detailed configuration and behaviors. You also can set the secure boot and control key management to prevent the unauthorized accessing for system.

► **Wake Up Event Setup sub-menu**

Sets system wake up behaviors for different sleep modes.

► **Secure Erase+**

Enables or disables Secure Erase+ function. **Secure Erase+** is the best way to effectively wipe all data from a SSD. Please note that data of SSD will be erased after enabling **Secure Erase+**.

► **Realtek PCIe GbE Family Controller (MAC sub-menu)**

This sub-menu shows driver information and configuration of the Ethernet controller parameter.

► **Boot sub-menu**

Use this menu to set the system boot states and the sequence of system boot devices.

► **Security sub-menu**

Use this menu to set the administrator password and the user password for system security. This menu also allows you to set the TPM (Trusted Platform Module) function.



Important

*When selecting the Administrator / User Password items, a password box will appear on the screen. Type the password then press **Enter**. The password typed now will replace any previous set password from CMOS memory. You will be prompted to confirm the password. You may also press **Esc** key to abort the selection.*

*To clear a set password, press **Enter** when you are prompted to enter a new password. A message will confirm the password is being disabled. Once the password is disabled, you can enter the setup and OS without authorization.*

► **Save & Exit**

This menu allows you to load the BIOS default values or factory default settings into the BIOS and exit the BIOS setup utility with or without changes.

OC Menu

This menu allows you to configure the frequencies and voltages for overclocking. Please note that, higher frequency and voltage may benefit overclocking capability but cause system un-stability.



Important

- *Overclocking your PC manually is only recommended for advanced users.*
- *Overclocking is not guaranteed, and if done improperly, it could void your warranty or severely damage your hardware.*
- *If you are unfamiliar with overclocking, we advise you to use **CREATOR GENIE** function for easy overclocking.*
- *The BIOS items in OC menu will vary with the processor.*

► **OC Explore Mode [Normal]**

Enables or disables to show the normal or expert version of OC settings.

[Normal] Provides the regular OC settings in BIOS setup.

[Expert] Provides the advanced OC settings for OC expert to configure in BIOS setup.

Note: We use * as the symbol for the OC settings of Expert mode.

► **CPU Ratio [Auto]**

Sets the CPU ratio that is used to determine CPU clock speed. This item can only be changed if the processor supports this function.

► **Advanced CPU Configuration**

Press **Enter** to enter the sub-menu. User can set the parameters about CPU power/current. The system may become unstable or unbootable after changing the parameters. If it occurs, please clear the CMOS data and restore the default settings.

► **A-XMP [Disabled]**

Please enable A-XMP or select a profile of memory module for overclocking the memory. This item will be available when the installed processor, memory modules and motherboard support this function.

► **DRAM Frequency [Auto]**

Sets the DRAM frequency. Please note the overclocking behavior is not guaranteed.

► **Adjusted DRAM Frequency**

Shows the adjusted DRAM frequency. Read-only.

‣ **FCLK Frequency [Auto]**

Sets the FCLK frequency (Internal Data Fabric clock of DRAM). Please note the overclocking behavior is not guaranteed.

‣ **UCLK DIV1 Mode [Auto]**

Sets UCLK (Internal memory controller clock) mode.

‣ **Memory Try It ! [Disabled]**

It can improve memory compatibility or performance by choosing optimized memory preset. This item will be available when the installed processor supports this function.

‣ **Memory Failure Retry [Enabled]**

Enables or disables the system reboot function when the memory OC retry fails.

‣ **Memory Failure Retry Count [2]**

Sets the count for memory OC retry. When memory OC retry fails reach the count, the memory will restore the last available settings. This item will display when the **Memory Failure Retry** sets to **Enabled**.

‣ **Advanced DRAM Configuration**

Press **Enter** to enter the sub-menu. User can set the memory timing for each/ all memory channel. The system may become unstable or unbootable after changing memory timing. If it occurs, please clear the CMOS data and restore the default settings. (Refer to the Clear CMOS jumper section to clear the CMOS data, and enter the BIOS to load the default settings.)

‣ **DigitALL Power sub-menu**

Press **Enter** to enter the sub-menu. In the sub-menu, you can setup some protecting conditions about voltage/ current/ tempature for CPU.

‣ **CPU Voltages control [Auto]**

These options allows you to set the voltages related to CPU. If set to **Auto**, BIOS will set these voltages automatically or you can set it manually.

‣ **DRAM Voltages control [Auto]**

These options allows you to set the voltages related to memory. If set to **Auto**, BIOS will set these voltages automatically or you can set it manually.

‣ **CHIP Voltages control [Auto] (optional)**

These options allows you to set the voltages related to chipset If set to **Auto**, BIOS will set these voltages automatically or you can set it manually.

► **Memory Changed Detect [Enabled]***

Enables or disables the system to issue a warning message during boot when the memory has been replaced.

[Enabled] The system will issue a warning message during boot and then you have to load the default settings for new devices.

[Disabled] Disables this function and keeps the current BIOS settings.

► **CPU Specifications sub-menu**

Press **Enter** to enter the sub-menu. This sub-menu displays the information of installed CPU. You can also access this information menu at any time by pressing [F4].
Read only.

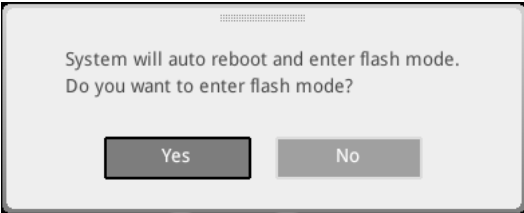
► **MEMORY-Z sub-menu**

Press **Enter** to enter the sub-menu. This sub-menu displays all the settings and timings of installed memory. You can also access this information menu at any time by pressing [F5].

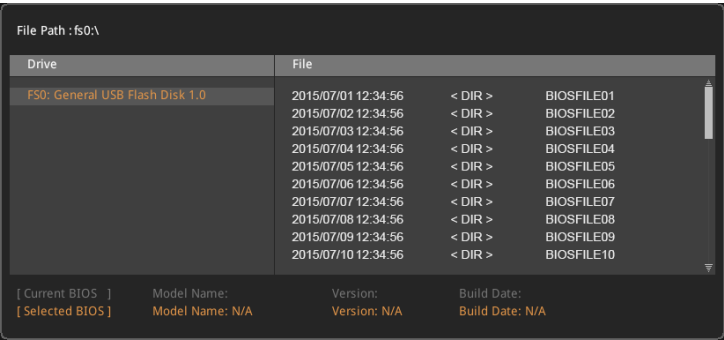
M-FLASH Menu

M-FLASH provides the way to update BIOS with a USB flash drive. Please download the latest BIOS file that matches your motherboard model from MSI website, save the BIOS file into your USB flash drive. And then follow the steps below to update BIOS.

1. Insert the USB flash drive that contains the update file into the computer.
2. Click on **M-FLASH** tab, a demand message will be prompted. Click on **Yes** to reboot and enter the flash mode.



3. The system will enter the flash mode and a file selection menu will appear after rebooting.



4. Select a BIOS file to perform the BIOS update process.
5. After the flashing process is 100% completed, the system will reboot automatically.

OC PROFILE Menu



► **Overclocking Profile 1/ 2/ 3/ 4/ 5/ 6**

Overclocking Profile 1/ 2/ 3/ 4/ 5/ 6 management. Press **Enter** to enter the sub-menu.

► **Set Name for Overclocking Profile 1/ 2/ 3/ 4/ 5/ 6**

Name the current overclocking profile.

► **Save Overclocking Profile 1/ 2/ 3/ 4/ 5/ 6**

Save the current overclocking profile.

► **Load Overclocking Profile 1/ 2/ 3/ 4/ 5/ 6**

Load the current overclocking profile.

► **Clear Overclocking Profile 1/ 2/ 3/ 4/ 5/ 6**

Clear the current overclocking profile.

► **OC Profile Load from ROM**

Load OC profile from BIOS ROM.

► **OC Profile Save to USB**

Save OC profile to the USB flash drive. The USB flash drive should be FAT/ FAT32 format only.

► **OC Profile Load from USB**

Load OC profile from the USB flash drive. The USB flash drive should be FAT/ FAT32 format only.

HARDWARE MONITOR Menu

This menu allows you to adjust the fan speed manually and monitor CPU/ system voltage.

Select a temperature curve line (white)
to be showed in Fan operating window

Select a fan mode for target fan

Select a fan to be configured

The screenshot shows the BIOS Hardware Monitor menu. It has tabs for Temperature, Fan Control, About, and Help. The Temperature tab is active, showing options for CPU Core, System, MIOs, CHIPSET, CPU Socket, and POIE. The Fan Control tab is also visible, showing options for CPU 1 (938RPM), PUMP 1 (0RPM), System 1 (0RPM), System 2 (0RPM), and System 3 (0RPM). Below these are radio buttons for PWM, DC, and Auto (selected). A 'Smart Fan Mode' checkbox is checked. A graph shows a temperature curve. At the bottom, there are buttons for 'All Full Speed(F)', 'All Set Default(D)', and 'All Set Cancel(C)'. A table at the very bottom displays various system temperatures and voltages.

Temperature	System	MIOs	CHIPSET	CPU socket	POIE
47°C	36°C	48°C	60°C	37°C	31°C
116°F	94°F	114°F	140°F	98°F	87°F

Voltage(V)	CPU Core	CPU NB/SOC	CPU VDDP	CPU 1P8	DRAM	CHIPSET Core	System 12V	System 5V
1.342	0.786	0.901	1.848	1.208	1.037	12.120	5.020	
3.300								

Click to enable the Smart Fan

Smart Fan duty information

Fan operating window

Setting Buttons

Temperature information

Voltage information

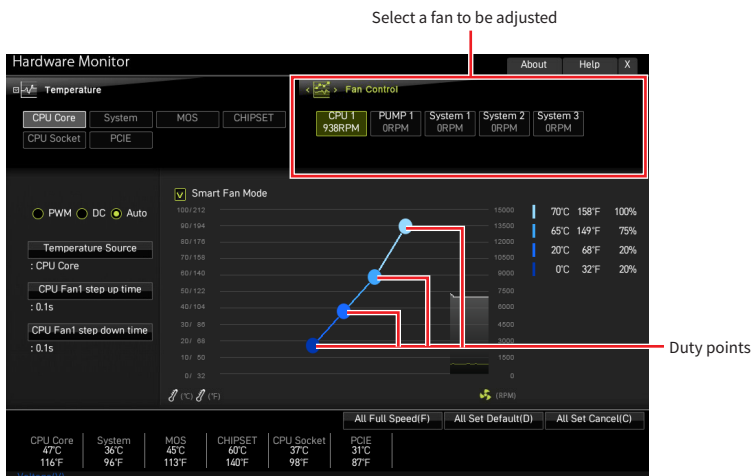
- **Smart Fan** - This setting enables/disables the Smart Fan function. Smart Fan is an excellent feature which will adjust the CPU/system fan speed automatically depending on the current CPU/system temperature, avoiding the overheating to damage your system.
- ▶ **Settings Buttons**
 - **All Full Speed** - configures all fans to run at full operating speed.
 - **All Set Default** - configures all fans' speeds return the BIOS default values.
 - **All Set Cancel** - discards current changes and restores previous settings for all fan .

Important

Make sure fans are working properly after adjusting the fan speed and switching the fan mode.

Adjusting fans

- 1. Selects a fan that you want to adjust and to display the fan duty curve line (yellow) in fan operating windows.
- 2. Click and drag the duty points to adjust the fan speed.



Important

The pictures in this section are for reference only and may vary from the motherboard you purchased.

AMD RAID Configuration

The following are the RAID levels supported by RAIDXpert2.

- RAID 0** (Striping) breaks the data into blocks which are written to separate hard drives. Spreading the hard drive I/O load across independent channels greatly improves I/O performance.
- RAID 1** (Mirroring) provides data redundancy by mirroring data between the hard drives and provides enhanced read performance.
- RAID 10** (Striped RAID1 Sets) uses four hard drives to create a combination of RAID 0 and 1 by forming a RAID 0 array from two RAID 1 arrays.
- Volume** (JBOD) provides the ability to link-together storage from one or several disks, regardless of the size of the space on those disks. Useful in scavenging space on disks unused by other disks in the array. Does not provide performance benefits or data redundancy.
- RAIDABLE** (also known as RAID Ready) allows the user to add more storage space or create a redundant array after a system is installed.

RAID level comparison

	RAID 0	RAID 1	RAID 10
Minimum # drives	2	2	4
Data protection	None	Excellent	Excellent
Read performance	Excellent	OK	OK
Write performance	Excellent	Good	Good
Capacity utilization	100%	50%	50%



All the information/ volumes/ pictures listed in your system might differ from the illustrations in this appendix.

Enabling RAIDXpert2 Configuration Utility

To enter the RAIDXpert2 Configuration Utility menu

- Power on and press **Delete** key to enter BIOS Setup menu.
- Press **F7** to switch to Advanced mode from EZ mode.
- Go to **BIOS > SETTINGS > Advanced > Integrated Peripherals > SATA Mode** and change setting to **RAID Mode**.
- Go to **BIOS > SETTINGS > Advanced > Windows OS Configuration > BIOS UEFI/ CSM Mode** and change setting to **UEFI**.
- Press **F10** to save configuration and exit, and then reboot and press **Delete** key to enter BIOS Setup menu.
- Go to **BIOS > SETTINGS > Advanced > RAIDXpert2 Configuration Utility** sub-menu.

Initializing Disks

New disks and legacy disks must be initialized before they can be used to create an AMD-RAID array. Initialization writes AMD-RAID configuration information (metadata) to a disk.

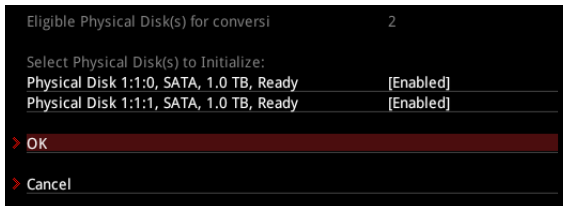


Important

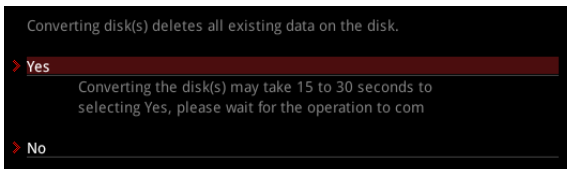
- If a disk is part of an AMD-RAID array, the disk cannot be selected for initialization. To initial the disk anyway, delete the AMD-RAID array. Data on the disk is deleted during initialization so ensure the correct disks are chosen to initialize.
- A legacy disk can contain valid data. When a legacy disk is initialized, all data on the disk is lost.

To initialize disks

1. As previously mentioned, enable RAIDXpert2 Configuration Utility.
2. Go to **BIOS > SETTINGS > Advanced > RAIDXpert2 Configuration Utility > Physical Disk Management > Select Physical Disk Operations > Initialize Disk** sub-menu.



3. Select desired disks by changing the Physical Disk setting to **Enabled**.
4. Select **OK**, then press Enter.



5. Review the warning message, if you want to proceed, select **YES**, then press Enter.
6. Initialization takes 10 to 15 seconds per disk. During initialization, a complete rescan of all channels is done automatically.

Creating Arrays

Arrays can be created after the disks are initialized.



Important

- For redundant arrays, the Create process is not started until after the operating system and AMD-RAID OS drivers have been installed and the system has booted to the operating system. However, the arrays are immediately available to use for either a bootable array or a data array.
- Array numbers are valid only for a given boot and might be different in the RAIDXpert2 Configuration Utility and RAIDXpert2. If a permanent label is required, use the RAIDXpert2 Web GUI Array Naming feature.
- At any point in the procedure, return to a prior window by pressing ESC.
- If the system is booted from an AMD-RAID bootable array, the first array in the Arrays section must be the bootable array. The system boots only from the first array in the Arrays section.

To create an array

1. As previously mentioned, enable RAIDXpert2 Configuration Utility.
2. Go to **BIOS > SETTINGS > Advanced > RAIDXpert2 Configuration Utility > Array Management > Create Array** sub-menu.

Select RAID Level: [RAID 1]

> Select Physical Disks

Configure Array Parameters:

Array Size: 0

Array Size Unit: []

Select CacheTagSize: [...]

Read Cache Policy: [Read Cache]

Write Cache Policy: [Write Back Ca...]

> Create Array

3. Select the RAID level from the **Select RAID Level** drop down menu.
4. Enter **Select Physical Disks** sub-menu, select member disks by changing the Physical Disk setting to **Enabled**.

Physical Disk 1:1:0, SATA, 1.0 TB, Ready [Enabled]

Physical Disk 1:1:1, SATA, 1.0 TB, Ready [Enabled]

Check All

Uncheck All

> Apply Changes

5. Select **Apply Changes**, then press Enter to apply and go back to previous sub-menu.
6. Change the **Select CacheTagSize**, **Read Cache Policy** and **Write Cache Policy** settings according to your needs.
7. Select **Create Array**, then press Enter.

Deleting Arrays

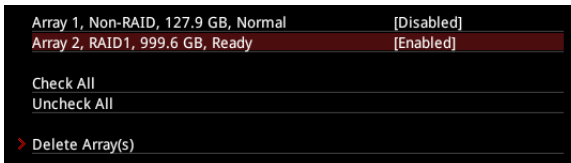


Important

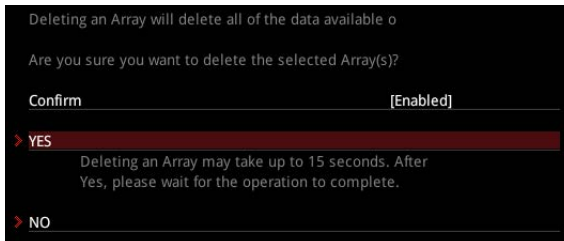
- Deleting an array permanently destroys all data that is on the array. This action cannot be undone and it is very unlikely that the data can be recovered.
- Do not delete the first array listed in the Arrays section, if it is the AMD-RAID bootable array. Doing this deletes the operating system and AMD-RAID files.

To delete an array

1. As previously mentioned, enable RAIDXpert2 Configuration Utility.
2. Go to **BIOS > SETTINGS > Advanced > RAIDXpert2 Configuration Utility > Array Management > Delete Array** sub-menu.



3. Select the desired array and change the setting to **Enabled**.
4. Enter **Delete Array(s)** sub-menu.



5. Review the warning message, if you want to proceed, Select **Confirm** and change the setting to **Enabled**.
6. Select **YES** then press Enter.

Installing RAID Driver

New Operating System Installation

The following details the installation of the drivers while installing operating system.

1. During the operating system installation, after selecting the location to install Windows click on **Load driver** button to install a third party RAID driver.
2. When prompted, insert the USB flash drive with **AMD RAID Drivers** and then click **Browse**.
 - To make an **AMD RAID Drivers** USB flash drive. Insert the MSI Driver Disc into the optical drive. Copy all the contents in \\Storage\AMD\
3. Navigate to the directory containing the saved AMD RAID drivers, then click **OK**.
4. Select the (**rcbottom.inf**) driver, click **Next**.
5. When prompted, click **OK**.
6. Click **Browse** and navigate to the directory containing the saved AMD RAID drivers again, then click **OK**.
7. Select the (**rcraid.inf**) driver, click **Next**.
8. You have successfully installed the RAID driver, and Windows setup should continue.
9. Leave the disk/ USB drive in the computer until the system reboots itself. Windows setup will need to copy the files after the RAID volume is formatted, and Windows setup starts copying files.

AMD RAIDXpert2 Management Suite Installation

1. Set the **SATA Mode** to **RAID Mode** in BIOS
2. Insert the MSI Driver Disc into the optical drive.
3. Click the **Select to choose what happens with this disc** pop-up notification, then select **Run DVDSetup.exe** to open the installer. If you turn off the AutoPlay feature from the Windows Control Panel, you can still manually execute the **DVDSetup.exe** from the root path of the MSI Driver Disc.
4. Under the **Drivers/Software** tab, check the **AMD RAID Drivers** check-box.
5. Click the **Install** button.
6. When prompt you to restart, click **OK** button to finish.
7. Restart your computer and enter the Windows operating system.
8. Double-click the **RAIDXpert2** icon to open the RAIDXpert2 Web GUI.
 - Default credentials are:
 - Username - **admin**
 - Password - **admin**
9. Change the credentials:
 - Create new username and password
10. Re-log into the RAIDXpert2 Web GUI with the new credentials.

Troubleshooting

Before sending the motherboard for RMA repair, try to go over troubleshooting guide first to see if your got similar symptoms as mentioned below.

The power is not on.

- Connect the AC power cord to an electrical outlet securely.
- Check if all ATX power connectors like **ATX_PWR1**, **CPU_PWR1** are connected from the power supply to the motherboard?
- Some power supply units have a power button on the rear side, make sure the button is turned on.
- Check if the power switch cable is connected to **JFP1** pin header properly.
- Verify the **Clear CMOS** jumper **JBAT1** is set to **Keep DATA**.
- Test with another known working power supply of equal or greater wattage.

The power is on, but no signal to monitor

- Connect the monitor power cord to a electrical outlet securely.
- Make sure the monitor is turned on.
- Select different inputs on the monitor.
- If 3 long beeps are heard, remove all memory modules and try to install only one memory module in the **DIMMA2** slot first and then restart the computer.
- If 1 long 2 short beeps are heard, remove and reinstall the graphics card and then restart the computer.
- Test with another known working graphics card.

The computer does not boot after updating the BIOS

- Clear the CMOS.
- Use the secondary BIOS to bootup the system (Only for motherboard with Dual BIOS)

Lost BIOS password

- Clear the CMOS, but that will cause you to lose all customized settings in the BIOS.

There is no audio

- Adjust the volume.
- Connect the speakers/headphones to audio ports on the motherboard rear IO panel.
- Remove secondary speakers/headphones, HDMI cables, USB audio devices.
- Test with another known working speaker or headphone.

There is no network

- Make sure the network chipset driver has been installed.
- Verify if the network cable is properly connected and make sure the LAN port LEDs are properly illuminated.
- Verify your TCP/IP settings.
- Restart or reset your router.
- Test with another known working LAN cable.

The USB device is not working

- Make sure your USB drive driver has been installed.
- Verify if USB device is listed in Windows® Device Manager.
- Connect the USB device to other USB port on the motherboard rear IO panel.

Regulatory Notices

FCC-B Radio Frequency Interference Statement

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and radiates radio frequency energy, and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:



- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.

NOTE

- The changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.
- Shield interface cables and AC power cord, if any, must be used in order to comply with the emission limits.

FCC Conditions

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:

- This device may not cause harmful interference.
- This device must accept any interference received, including interference that may cause undesired operation.

MSI Computer Corp.

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CE Conformity

Products bearing the CE marking comply with one or more of the following EU Directives as may be applicable:

- RED 2014/53/EU
- Low Voltage Directive 2014/35/EU
- EMC Directive 2014/30/EU
- RoHS Directive 2011/65/EU
- Implementing measure Directive 2009/125/EC of ESPR Regulation (EU) 2024/1781



Compliance with these directives is assessed using applicable European Harmonized Standards.

For any support regarding the EU General Product Safety Regulation (GPSR), please contact MSI Computer Europe B.V. via <https://msi.gm/gpsr> Churchilllaan 202, 5705 BK Helmond, the Netherlands.

Products with Radio Functionality (EMF)

This product incorporates a radio transmitting and receiving device. For computers in normal use, a separation distance of 20 cm ensures that radio frequency exposure levels comply with EU requirements. Products designed to be operated at closer proximities, such as tablet computers, comply with applicable EU requirements in typical operating positions. Products can be operated without maintaining a separation distance unless otherwise indicated in instructions specific to the product.

Restrictions for Products with Radio Functionality



WLAN 5150-5350 MHz:

The WiFi operating in the band 5150-5350 MHz shall be restricted to indoor use in the European Union.

WLAN 6E:

a. Low Power Indoor (LPI) Wi-Fi 5.945-6.425 GHz devices:

The device is restricted to indoor use only when operating in the 5945 to 6425 MHz frequency range in the European Union.

b. Very Low Power (VLP) Wi-Fi 5.945-6.425 GHz devices (portable devices):

The device is not permitted to be used on Unmanned Aircraft Systems (UAS) when operating in the 5945 to 6425 MHz frequency range in the European Union.

Wireless Radio Use

This device is restricted to indoor use when operating in the 2.4GHz, 5GHz, 6GHz frequency band.

Cet appareil doit être utilisé à l'intérieur.

당해 무선설비는 운용중 전파혼신 가능성이 있음.

この製品は、周波数帯域 2.4GHz, 5GHz, 6GHz で動作しているときは、屋内においてのみ使用可能です。

NCC無線設備警告聲明

工作頻率2.4GHz, 5GHz, 6GHz該頻段限於室內使用。

經型式認證合格之低功率射頻電機，非經許可，公司、商號或使用者均不得擅自變更頻率、加大功率或變更原設計之特性及功能。

低功率射頻電機之使用不得影響飛航安全及干擾合法通信；經發現有干擾現象時，應立即停用，並改善至無干擾時方得繼續使用。前項合法通信，指依電信法規定作業之無線電通信。低功率射頻電機須忍受合法通信或工業、科學及醫療用電波輻射性電機設備之干擾。

Compliance Statement of Innovation, Science and Economic Development Canada (ISED)

This device complies with with Innovation, Science and Economic Development Canada's licence-exempt RSS(s). Operation is subject to the following two conditions:

(1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

Operation in the band 5150-5250 MHz is only for indoor use to reduce the potential for harmful interference to co-channel mobile satellite systems.

CAN ICES-003(B)/NMB-003(B)

Australia and New Zealand notice

This equipment incorporates a radio transmitting and receiving device. In normal use, a separation distance of 20 cm ensures that radio frequency exposure levels comply with the Australian and New Zealand Standards.

クラスB情報技術装置

この装置は、クラスB情報技術装置です。この装置は、家庭環境で使用することを目的としていますが、この装置がラジオやテレビジョン受信機に近接して使用されると、受信障害を引き起こすことがあります。取扱説明書に従って正しい取り扱いをして下さい。

VCCI-B

KC인증서



R-R-MSI-10-7C95

상호: (주)엠에스아이코리아

제품명: 메인보드

모델명: 10-7C95

제조년월: 2020년

제조사 및 제조국가: MSI/중국



Battery Information

European Union:



Batteries, battery packs, and accumulators should not be disposed of as unsorted household waste. Please use the public collection system to return, recycle, or treat them in compliance with the local regulations.

BSMI:



廢電池請回收

For better environmental protection, waste batteries should be collected separately for recycling or special disposal.

California, USA:



The button cell battery may contain perchlorate material and requires special handling when recycled or disposed of in California.

For further information please visit:

<https://dtsc.ca.gov/perchlorate/>

⚠ WARNING

- **INGESTION HAZARD:** This product contains a button cell or coin battery.
- **DEATH** or serious injury can occur if ingested.
- A swallowed button cell or coin battery can cause **Internal Chemical Burns** in as little as **2 hours**.
- **KEEP** new and used batteries **OUT OF REACH OF CHILDREN**
- **Seek immediate medical attention** if a battery is suspected to be swallowed or inserted inside any part of the body.



- Remove and immediately recycle or dispose of used batteries according to local regulations and keep away from children. Do NOT dispose of batteries in household trash or incinerate.
- Even used batteries may cause severe injury or death. Call a local poison control center for treatment information.
- Battery type: CR2032
- Battery voltage: 3V
- Non-rechargeable batteries are not to be recharged.
- Do not force discharge, recharge, disassemble, heat above (manufacturer's specified temperature rating) or incinerate. Doing so may result in injury due to venting, leakage or explosion resulting in chemical burns.
- This product contains an irreplaceable battery.
- This icon indicates that a swallowed button battery can cause serious injury or death. Please keep batteries out of sight or reach of children.

AVERTISSEMENT

- **RISQUE D' INGESTION:** ce produit contient une pile bouton.
- Si elle est ingérée, elle peut entraîner la **MORT** ou provoquer de graves blessures.
- Une pile bouton ingérée peut provoquer des **brûlures chimiques internes** en seulement **2 heures**.
- **GARDER** les piles neuves et usagées **HORS DE PORTÉE des ENFANTS**.
- **Demander une aide médicale immédiate** en cas de suspicion d' ingestion ou d' insertion d' une pile dans une partie quelconque du corps.



- Retirer les piles usagées et procéder immédiatement au recyclage ou à la mise au rebut conformément aux réglementations locales, et les tenir hors de portée des enfants. Ne PAS jeter les piles aux ordures ménagères ni les incinérer.
- Les piles même usagées peuvent provoquer des blessures graves, voire mortelles. Appeler un centre antipoison local pour obtenir des informations sur le traitement.
- Type de pile: CR2032
- Tension nominale de la batterie: 3V
- Les piles non rechargeables ne doivent pas être rechargées.
- Ne pas forcer la décharge, recharger, démonter, chauffer au-delà de (la température spécifiée par le fabricant) ou incinérer. Le non-respect de cette instruction pourrait entraîner des blessures dues à une fuite d' air ou de liquide ou à une explosion occasionnant des brûlures chimiques.
- Ce produit contient une batterie non remplaçable.
- Cet icône indique qu' une pile bouton avalée peut causer des blessures graves, voire mortelles. Veuillez conserver les piles hors de vue et de portée des enfants.



Chemical Substances Information

In compliance with chemical substances regulations, such as the EU REACH Regulation (Regulation EC No. 1907/2006 of the European Parliament and the Council), MSI provides the information of chemical substances in products at:

<https://csr.msi.com/global/index>

Environmental Policy

- The product has been designed to enable proper reuse of parts and recycling and should not be thrown away at its end of life.
- Users should contact the local authorized point of collection for recycling and disposing of their end-of-life products.
- Visit the MSI website <https://csr.msi.com/global/pevn_ewaste> and locate a nearby distributor for further recycling information.
- Please visit <<https://us.msi.com/page/recycling>> for information regarding the recycling of your product in the US.

WEEE (Waste Electrical and Electronic Equipment) Statement



European Union: This symbol on the product indicates that this product cannot be discarded as municipal waste. Instead, it is your responsibility to dispose of your waste electrical and electronic equipment by handing it over to a designated collection point for recycling. For more information about where you can drop off your waste equipment for recycling, please contact your local city office, your household waste disposal service or the shop where you purchased the product.

日本JIS C 0950材質宣言

日本工業規格JIS C 0950により、2006年7月1日以降に販売される特定分野の電気および電子機器について、製造者による含有物質の表示が義務付けられます。

<https://csr.msi.com/tw/Japan-JIS-C-0950-Material-Declarations>

India RoHS

This product complies with the “India E-waste (Management and Handling) Rule 2011” and prohibits use of lead, mercury, hexavalent chromium, polybrominated biphenyls or polybrominated diphenyl ethers in concentrations exceeding 0.1 weight % and 0.01 weight % for cadmium, except for the exemptions set in Schedule 2 of the Rule.

Türkiye EEE yönetmeliği

Türkiye Cumhuriyeti: EEE Yönetmeliğine Uygundur

Україна обмеження на наявність небезпечних речовин

Обладнання відповідає вимогам Технічного регламенту щодо обмеження використання деяких небезпечних речовин в електричному та електронному обладнанні, затвердженого постановою Кабінету Міністрів України від 10 березня 2017 р. № 139

Việt Nam RoHS

Kể từ ngày 01/12/2012, tất cả các sản phẩm do công ty MSI sản xuất tuân thủ Thông tư số 30/2011/TT-BCT quy định tạm thời về giới hạn hàm lượng cho phép của một số hóa chất độc hại có trong các sản phẩm điện, điện tử”

Bangladesh RoHS Statement

This product shipping to Bangladesh comply with RoHS requirements as prescribed in the Hazardous Waste (E-waste) Management Rules, 2021 by Government of Bangladesh.

For more information, please refer:

<https://csr.msi.com/global/bangladesh-rohs-statement>

Bangladesh E-waste Management Reminder

Please manage, dismantle, dispose of, or recycle e-waste properly as per the Environmental Protection Rules, 1997 to avoid the harm that may be caused to the environment and ecosystem, public interest and public health.

Material Declaration under India’ s Plastic Waste Management Rules:

INDOMSI COMPUTERS PRIVATE LIMITED

Regn. No.: IM-06-000-06-AAHCI0478A-25

Thickness $\geq$ 50 microns

MS-7C95主板产品中有害物质的名称及含有信息表

部件名称	有害物质									
	Pb	Hg	Cd	Cr(VI)	PBBs	PBDEs	DBP	DIBP	BBP	DEHP
电路板组件*	×	○	○	○	○	○	○	○	○	○
外部信号连接头	×	○	○	○	○	○	○	○	○	○
线材	×	○	○	○	○	○	○	○	○	○
注1：○：表示该有害物质在该部件所有均质材料中的含量均不超出电器电子产品有害物质限制使用国家标准要求。 ×：表示该有害物质至少在该部件的某一均质材料中的含量超出电器电子产品有害物质限制使用国家标准要求。 注2：以上未列出的部件，表明其有害物质含量均不超出电器电子产品有害物质限制使用国家标准要求。 注3：上述表格标注“×”之部件，皆符合达标管理目录限用物质应用例外清单之限值要求。 * 电路板组件：包括印刷电路板及其构成的零部件。										

限用物質含有情況標示聲明書

設備名稱:電腦主機板			型號(型式):MS-7C95			
單元	限用物質及其化學符號					
	鉛 (Pb)	汞 (Hg)	鎘 (Cd)	六價鉻 (Cr ⁺⁶)	多溴聯苯 (PBB)	多溴二苯醚 (PBDE)
電路板	○	○	○	○	○	○
電子元件	—	○	○	○	○	○
金屬機構件	—	○	○	○	○	○
塑膠機構件	○	○	○	○	○	○
備考1. “超出0.1 wt %”及“超出0.01 wt %”係指限用物質之百分比含量超出百分比含量基準值。						
備考2. “○”係指該項限用物質之百分比含量未超出百分比含量基準值。						
備考3. “—”係指該項限用物質為排除項目。						

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HDMI™
HIGH-DEFINITION MULTIMEDIA INTERFACE

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Technical Support

If a problem arises with your system and no solution can be obtained from the user guide, please contact your place of purchase or local distributor. Alternatively, please try the following help resources for further guidance.

- Visit the MSI website for technical guide, BIOS updates, driver updates, and other information: <https://www.msi.com/support/>
- Register your product at: <http://register.msi.com>

Revision History

- Version 1.0, 2020/06, First release.
- Version 1.4, 2020/10, updated release.
- Version 1.7, 2021/04, add B560M PRO-VDH WIFI6.
- Version 1.8, 2023/10, update Wi-Fi information.
- Version 1.9, 2024/01, update package contents.
- Version 1.A, 2024/03, Add battery safety warning.
- Version 1.C, 2026/07, update Wi-Fi information.

